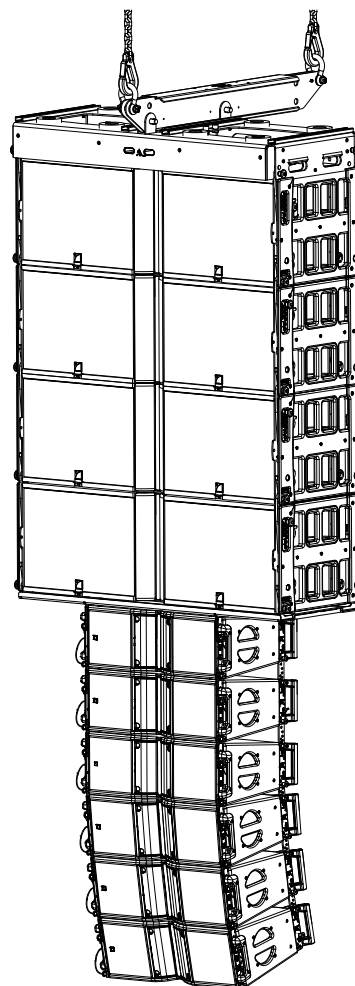
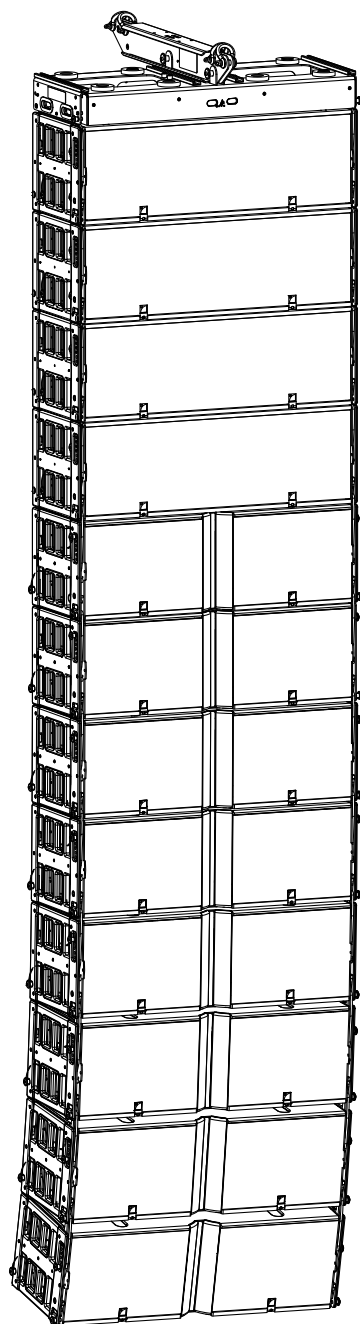


K 1



owner's manual (EN)



Document reference: K1 owner's manual (EN) version 2.0

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Safety

Instructions



Inspect the system before any deployment.

Perform safety related checks and inspections before any deployment.

Perform preventive maintenance at least once a year.

Refer to the preventive maintenance section for a list of actions and their periodicity.

Insufficient upkeep of the product can void the warranty.

If any safety issue is detected during inspection, do not use the product before performing corrective maintenance.

Check for issues. A rigging system part or fastener is missing or loose. A rigging system part exhibits: bends, breaks, broken parts, corrosion, cracks, cracks in welded joints, deformation, denting, wear, holes. A safety cue or label is missing.



Never incorporate equipment or accessories not approved by L-Acoustics.

Read all the related PRODUCT INFORMATION documents shipped with the products before exploiting the system.



Do not store the product on an unstable cart, stand, tripod, bracket, or table.



Beware of sound levels.

Do not stay within close proximity of loudspeakers in operation.

Loudspeaker systems are capable of producing very high sound pressure levels (SPL) which can instantaneously lead to permanent hearing damage to performers, production crew, and audience members. Hearing damage can also occur at moderate level with prolonged exposure to sound.

Check the applicable laws and regulations relating to maximum sound levels and exposure times.



Work with qualified personnel for rigging the system.

Installation should only be carried out by qualified personnel that are familiar with the rigging techniques and safety recommendations outlined in this manual.

Ensure personnel health and safety.

During installation and set-up personnel must wear protective headgear and footwear at all times. Under no circumstances is personnel allowed to climb on a loudspeaker assembly.

Respect the Working Load Limit (WLL) of third party equipment.

L-Acoustics is not responsible for any rigging equipment and accessories provided by third party manufacturers.

Verify that the Working Load Limit (WLL) of the suspension points, chain hoists and all additional hardware rigging accessories is respected.

Respect the maximum configurations and the recommended safety precautions.

For safety issue, respect the maximum configurations outlined in this manual. To check the conformity of any configuration in regards with the safety precautions recommended by L-Acoustics, model the system in Soundvision and refer to the warnings in Mechanical Data section.

Be cautious when flying a loudspeaker configuration.

Before installing/raising the product, check each individual element to make sure that it is securely fastened to the adjacent element. Always verify that no one is standing underneath the product when it is being installed/raised. Never leave the product unattended during the installation process.

As a general rule, L-Acoustics recommends the use of secondary safety at all times.

Be cautious when ground-stacking a loudspeaker array.

Do not stack the loudspeaker array on unstable ground or surface. If the array is stacked on a structure, platform, or stage, always check that the latter can support the total weight of the array.

As a general rule, L-Acoustics recommends the use of safety straps at all times.

Risk of falling objects

Verify that no unattached items remain on the product or assembly.

Risk of tipping

Remove all rigging accessories before transporting a product or an assembly.

Take into account the wind effects on dynamic load.

When a loudspeaker assembly is deployed in an open air environment, wind can produce dynamic stress to the rigging components and suspension points.

If the wind force exceeds 6 bft (Beaufort scale), lower down and/or secure the product or the assembly.

**Intended use**

This system is intended for use by trained personnel for professional applications.



As part of a continuous evolution of techniques and standards, L-Acoustics reserves the right to change the specifications of its products and the content of its documents without prior notice.

Check www.l-acoustics.com on a regular basis to download the latest document and software updates.

**Long term exposure to extreme conditions may damage the product.**

For more information, refer to the **Products weather protection** document, available on the website.

**Read the maintenance section of this document before servicing the product.****Contact L-Acoustics for advanced maintenance.**

Any unauthorized maintenance operation will void the product warranty.



This marking indicates that this product should not be disposed of with other household waste throughout the EU. To prevent possible harm to the environment or human health from uncontrolled waste disposal, recycle it responsibly to promote the sustainable reuse of material resources. To return your used device, please use the return and collection systems or contact the retailer where the product was purchased. They can take this product for environmentally safe recycling.



Introduction

K1 large format WST line source



The K1 enclosure is the full-range element of a large format WST® line source with variable curvature. The K1 features two 15" speakers mounted in a bass-reflex enclosure, four 6.5" high-efficiency direct radiating speakers, and three 3" diaphragm compression drivers coupled to individual DOSC® waveguides. The transducers are implemented in a K-shape configuration. The K1 enclosure is based on a 3-way active design with 2 LF sections, 1 MF section and 1 HF section, each one at the nominal impedance of 8 ohms. The K1 enclosure is made of first grade Baltic birch plywood to ensure maximum acoustical and mechanical integrity. A four-point rigging system is integrated into the cabinet.

The K1 enclosure operates over the nominal frequency range of 35 Hz to 20 kHz. The frequency response of the system can be extended down to either 30 Hz or 25 Hz with the addition of the dedicated K1-SB extension, or the KS28 subwoofer, respectively. The K1 generates a symmetric horizontal coverage of 90°.

The rigging system allows vertically assembling K1 enclosures with various inter-element angles up to 5°, constituting a line array with variable curvature. The combination of the coplanar symmetry and the DOSC® waveguide in the HF region ensure a perfect acoustic coupling between the elements of an array. The WST® (Wavefront Sculpture Technology) criteria are fulfilled, so that such an array can be qualified as a true line source. Any WST® line source provides a smooth tonal response and a coverage that is free of secondary lobes over the entire frequency range.

The K1 system is driven by the LA12X amplified controllers which ensure active system linearization, intelligent transducer protection, and optimization of the loudspeaker system in its different operating modes. The acoustic performances of the system will depend upon the used preset and the chosen physical configuration

How to use this manual

The K1 owner's manual is intended for all actors involved in the system design, implementation, preventive and corrective maintenance of the K1 system. It must be used as follows:

1. Read the technical description for an overview of all system elements, their features, and their compatibilities.
 - [Electro-acoustical description](#) (p.13)
 - [Rigging system description](#) (p.17)
2. Prepare the system configuration. Consider the mechanical limits and the available acoustical configurations.
 - [Mechanical safety](#) (p.28)
 - [Loudspeaker configurations](#) (p.30)
3. Before rigging the system, perform mandatory inspections and functional checks.
 - [Inspection and preventive maintenance](#) (p.41)
4. To deploy the system, follow the step-by-step rigging instructions and refer to the cabling schemes.
 - [Rigging procedures](#) (p.62)
 - [Connection to LA amplified controllers](#) (p.104)



The [Corrective maintenance](#) (p.106) section contains the operations authorized for the end user. Performing another operation exposes to hazardous situations. For advanced maintenance, contact your L-Acoustics representative.

As part of a continuous evolution of techniques and standards, L-Acoustics reserves the right to change the specifications of its products and the content of its document without prior notice. Please check www.l-acoustics.com on a regular basis to download the latest document and software updates.

Contact information

For information on advanced corrective maintenance:

- contact your Certified Provider or your L-Acoustics representative
- for Certified Providers, contact the L-Acoustics customer service: customer.service@l-acoustics.com (EMEA/APAC), laus.service@l-acoustics.com (Americas).

Symbols

The following symbols are used in this document:



This symbol indicates a potential risk of harm to an individual or damage to the product.

It can also notify the user about instructions that must be strictly followed to ensure safe installation or operation of the product.



This symbol notifies the user about instructions that must be strictly followed to ensure proper installation or operation of the product.



This symbol notifies the user about complementary information or optional instructions.

Revision history

version number	publication date	modification
1.0	Jun. 2024	Initial version.
2.0	Aug. 2024	• Updated K1-DELTA (p.22) • Updated Corrective maintenance (p.106). • Updated Specifications (p.119)

System components

Loudspeaker enclosures

K1	3-way full-range active WST® enclosure
K1-SB	K1 system subwoofer 2×15"
K2	3-way full-range active WST enclosure
Kara II	2-way active WST® enclosure: 2 × 8" LF + 3" HF diaphragm
KS28	Flyable subwoofer 2 × 18"

Powering and driving system

LA12X	Amplified controller 4 × 2600 W / 4 ohms
-------	--



Refer to the LA12X owner's manual for operating instructions.

Racks

LA-RAK II AVB	Touring rack containing three LA12X, LA-POWER II for power distribution, LA-PANEL II for audio and network distribution, and two LS10 for AVB distribution
---------------	--

Cables

SP cables	SpeakON loudspeaker cables (4-point, 4 mm ² gauge) Come in different sizes: SP.7 (0.7 m / 2.3 ft), SP5 (5 m / 16.4 ft), SP10 (10 m / 32.8 ft), and SP25 (25 m / 82 ft)
SP-Y1	Breakout cable for two passive enclosures (2.5 mm ² gauge) 4-point speakON to 2 × 2-point speakON, provided with a CC4FP adapter
DO cables	PA-COM loudspeaker cables (8-point, 4 mm ² gauge) Come in different sizes: DO.7 (0.7 m / 2.3 ft), DO.10 (10 m / 32.8 ft), and DO.25 (25 m / 82 ft)
DOSUB-LA8	Breakout cable for four passive enclosures (4 mm ² gauge) 8-point PA-COM to 4 × 2-point speakON
DOFILL-LA8	Breakout cable for two 2-way active enclosures (4 mm ² gauge) 8-point PA-COM to 2 × 4-point speakON
DO3WFILL	Breakout cable for one 2-way active enclosure and two passive enclosures (4 mm ² gauge) 8-point PA-COM to 1 × 4-point speakON and 2 × 2-point speakON



Information about the connection of the enclosures to the LA amplified controllers is given in this document.

Refer to the LA12X owner's manual for detailed instructions about the whole cabling scheme, including modulation cables and network.

Rigging elements

K1-BUMP	Structure for flying and stacking K1 and K1-SB arrays
K1-DELTA	Rigging accessory for rear attachment of 2 motors to K1-BUMP
KARA-DOWNK1	Flying bumper for rigging KARA (II) under K1 or K1-SB
K1-LASERMOUNT	K1 laser support plate (compatible TEQSAS / SSE Prosight / Align Array / KSG)
K2-LINK	Rigging accessory for rear attachment of K2 below K1

Transportation accessories

K-BUMPFLIGHT	Modular flight case for 2 K1-BUMP or 2 K2-BUMP
K1-CHARIOT	Chariot for 4 × K1 or 4 × K1-SB
K1-CHARIOTCOV	Protective cover for 4 × K1/K1-SB on CHARIOT
K1-PLA	Removable front dolly for K1 or K1-SB enclosure
K1-COV	Protective cover for K1 enclosure
K1-SBCOV	Protective cover for K1-SB enclosure

Software applications

Soundvision	3D acoustical and mechanical modeling software
LA Network Manager	Software for remote control and monitoring of amplified controllers

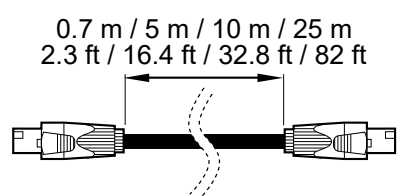


Refer to the Soundvision help.

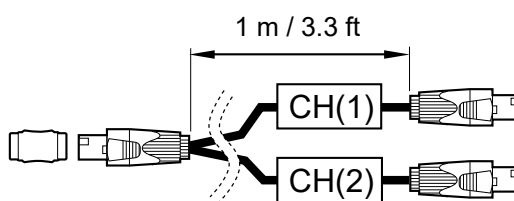
Refer to the LA Network Manager help.

System component illustrations

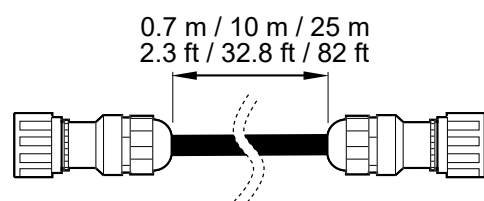
Cables



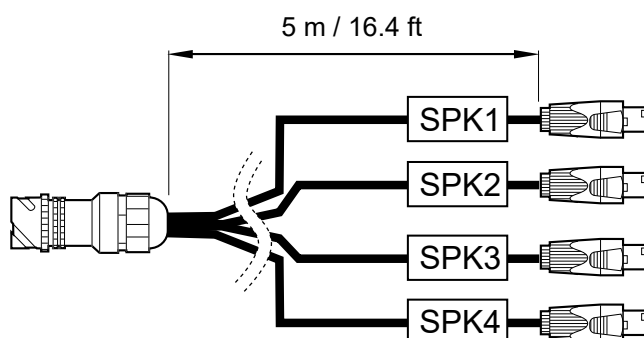
SP cables



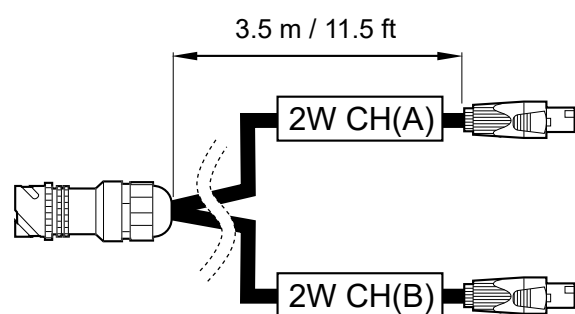
SP-Y1



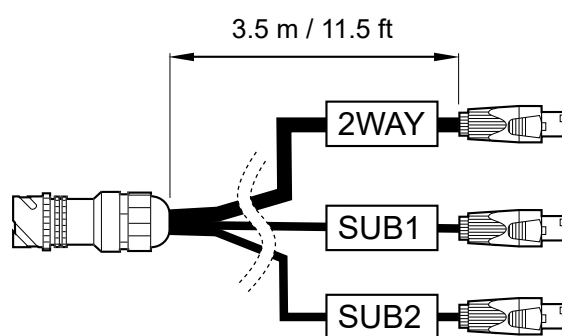
DO cables



DOSUB-LA8

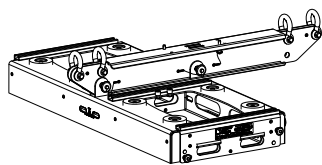


DOFILL-LA8

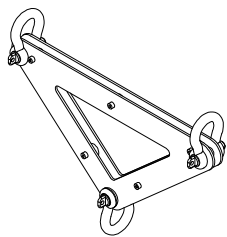


DO3WFILL

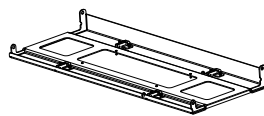
Rigging accessories



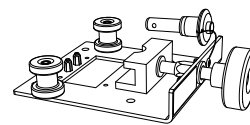
K1-BUMP



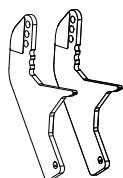
K1-DELTA



KARA-DOWNK1

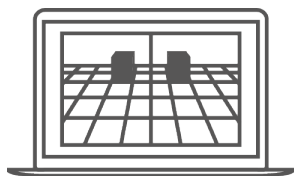


K1-LASERMOUNT



K2-LINK

Software applications

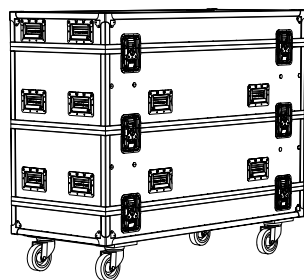


Soundvision

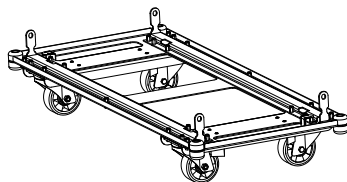


LA Network Manager

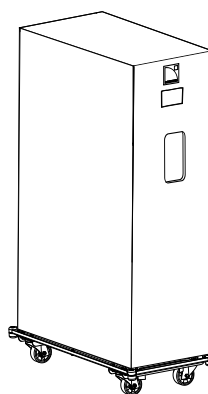
Transportation



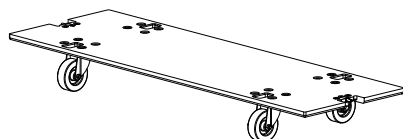
K-BUMPFLIGHT



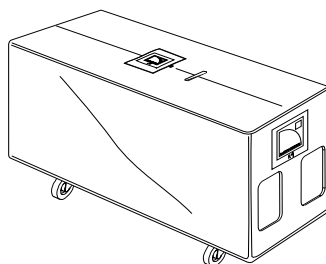
K1-CHARIOT



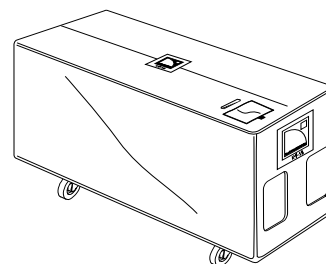
K1-CHARIOTCOV



K1-PLA



K1-COV

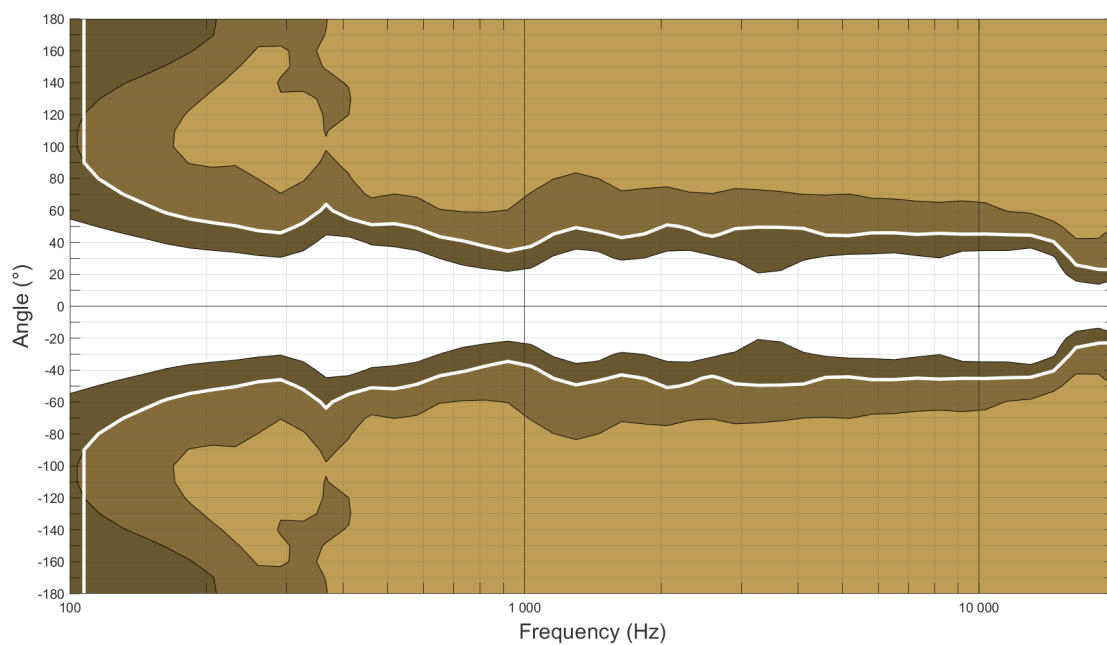
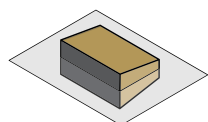


K1-SBCOV

Electro-acoustical description

Directivity

Horizontal



Dispersion angle diagram of a K1 array, using lines of equal sound pressure at -3 dB, -6 dB, -12 dB.

Preset description

[K1] [K2 70] [K2 90] [K2 110]

loudspeaker elements	outputs	channels	routing	gain	delay	polarity	mute
left LF	OUT 1	LF	IN A	0 dB	0 ms	+	ON
right LF	OUT 2	LF					ON
MF	OUT 3	MF					ON
HF	OUT 4	HF					ON

[KARAIIDOWNK1] [KARAIIDOWNK1 70] [KARAIIDOWNK1 90]

loudspeaker elements	outputs	channels	routing	gain	delay	polarity	mute
LF	OUT 1	LF	IN A	0 dB	0 ms	+	ON
HF	OUT 2	HF					ON
LF	OUT 3	LF	IN A	0 dB	0 ms	+	ON
HF	OUT 4	HF					ON

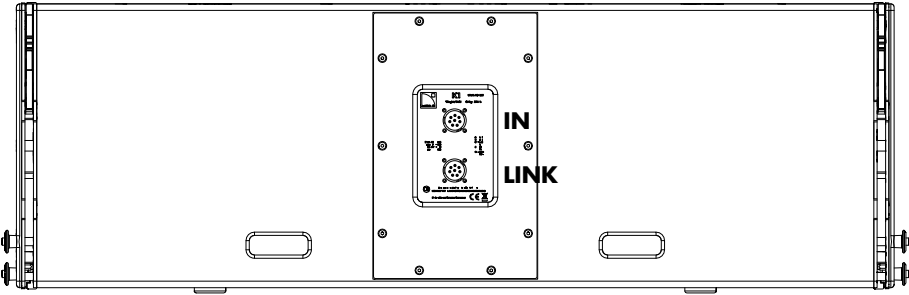
[K1SB_60] [K1SB_X] [KS28_60]

outputs	channels	routing	gain	delay	polarity	mute
OUT 1	SB	IN A	0 dB	0 ms	+	ON
OUT 2	SB	IN A	0 dB	0 ms	+	ON
OUT 3	SB	IN A	0 dB	0 ms	+	ON
OUT 4	SB	IN A	0 dB	0 ms	+	ON

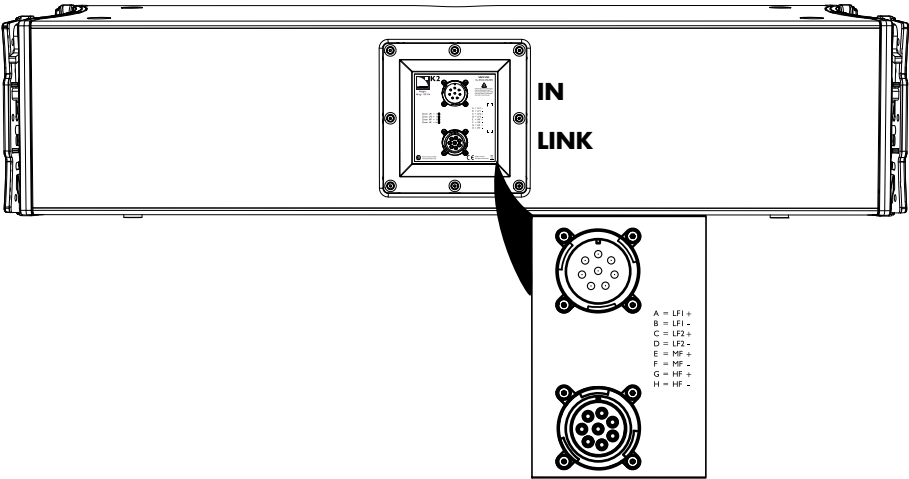
[KS28_60_C] [KS28_60_Cx]

loudspeaker elements	outputs	channels	routing	gain	delay	polarity	mute
SR	OUT 1	SR	IN A	0 dB	0 ms	+	ON
SB	OUT 2	SB					ON
SB	OUT 3	SB					ON
SB	OUT 4	SB					ON

Connectors



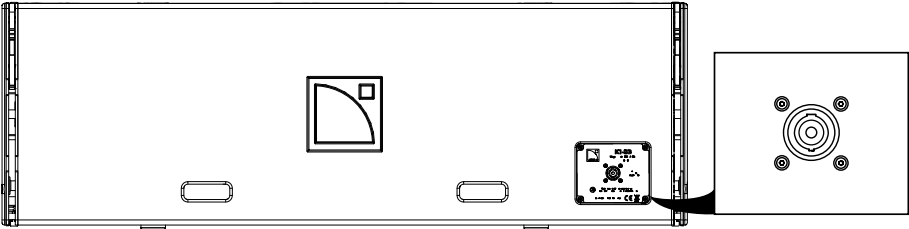
K1
2 x 8-point PA-COM



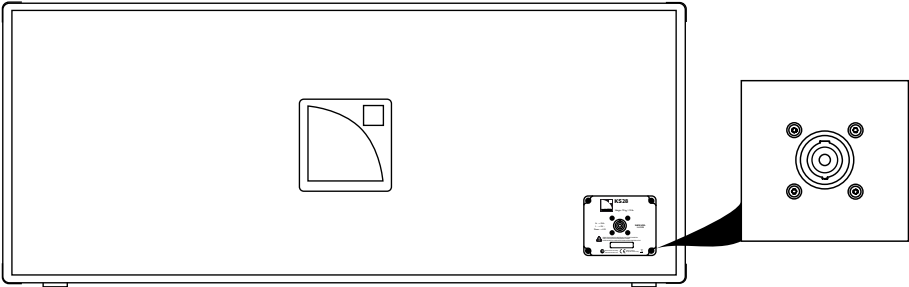
K2
2 x 8-point PA-COM

Internal pinout for L-Acoustics 3-way active enclosures

PA-COM points	A/B	C/D	E/F	G/H
Transducer connectors	left LF	right LF	MF	HF



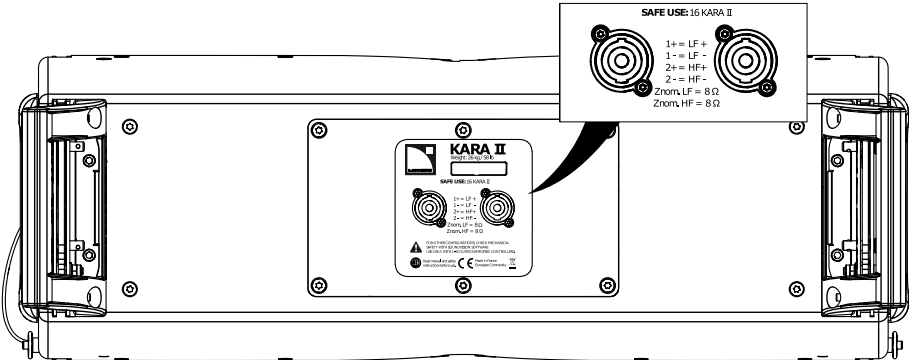
K1-SB
1 x 4-point speakON



KS28
1 x 4-point speakON

Internal pinout for L-Acoustics subwoofers

SpeakON points	1 +	1 -	2 +	2 -
Transducer connectors	LF +	LF -	Not linked	Not linked



Kara II
2 x 4-point speakON

Internal pinout for L-Acoustics 2-way active enclosures

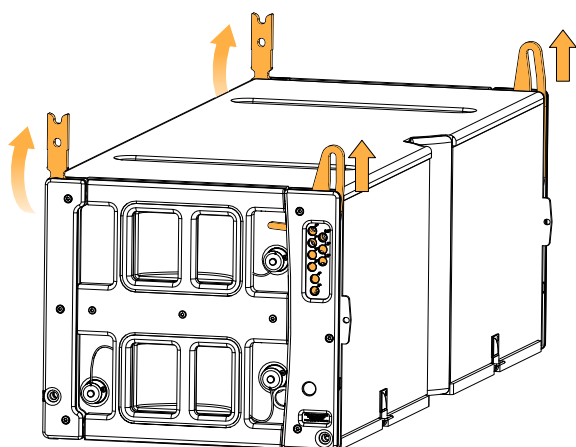
SpeakON points	1 +	1 -	2 +	2 -
Transducer connectors	LF +	LF -	HF +	HF -

Rigging system description

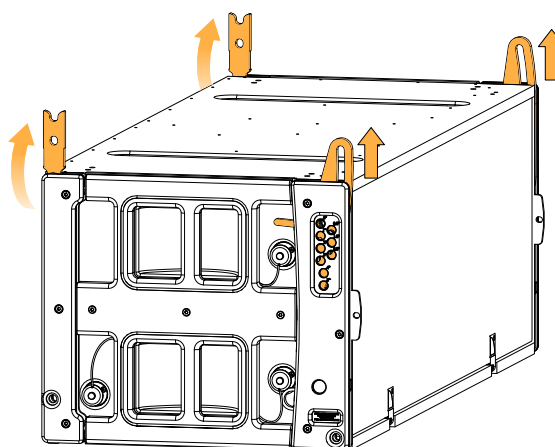
K1/K1-SB

On both sides, K1/K1-SB integrates two arms to connect another element of the rigging system, such as an enclosure or to a flying frame:

- At the front, a sliding arm enables the inter-element angle setting and the inter-element front connection. Nine angle positions are available: 0°, 0.5°, 1°, 1.5°, 2°, 2.5°, 3°, 4°, 5°.
- At the rear, a rotating arm enables the inter-element rear connection.

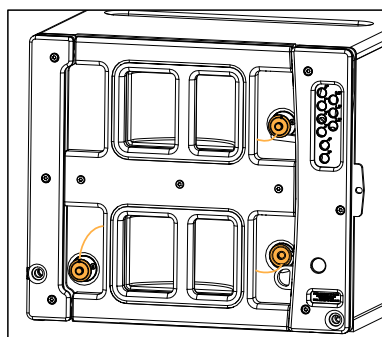


K1

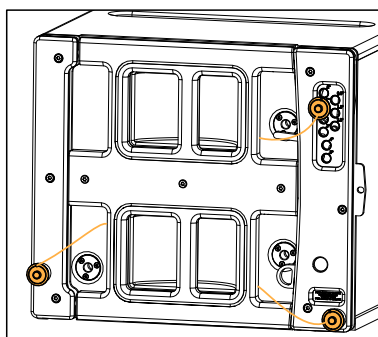


K1-SB

The rigging arms are secured with three ball-locking pins on each side of the enclosure:



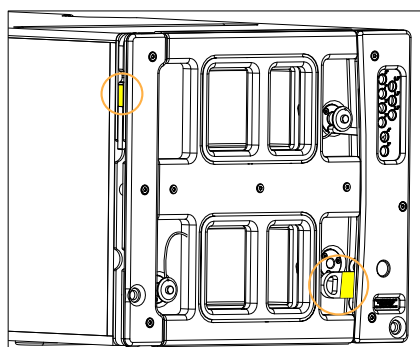
Ball-locking pins in storage position



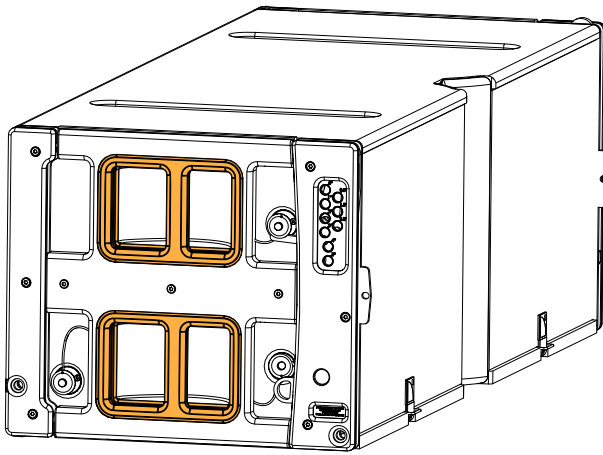
Ball-locking pins in rigging position

Adjacent enclosures or rigging accessories are secured by an automatic system with buttons and latches. The system is activated in advance and locks itself during the stacking and lifting procedures.

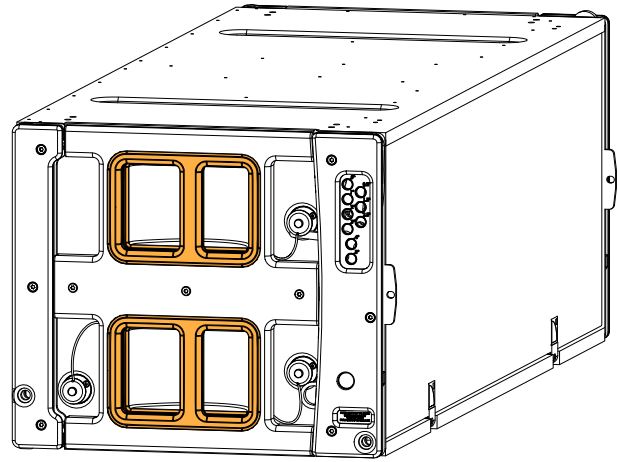
The rigging elements and latches are fitted with yellow safety labels that are visible when they are not safely locked.



Two ergonomic handles on each side of K1 are available for handling.

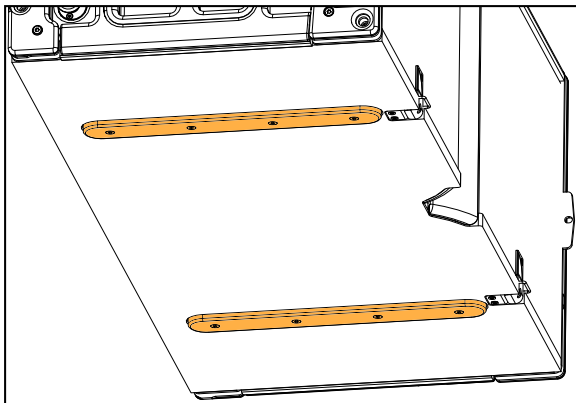


K1

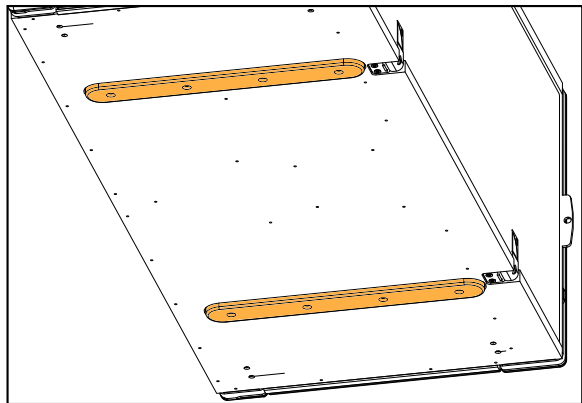


K1-SB

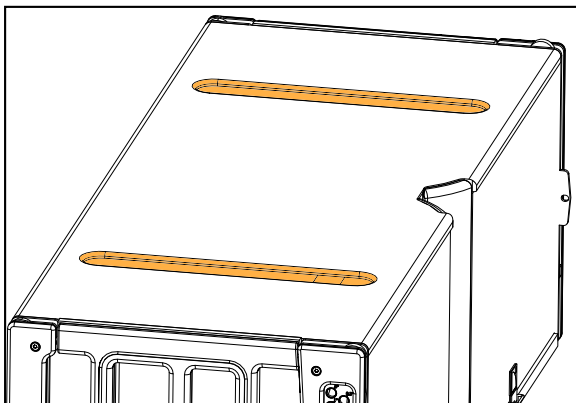
K1/K1-SB features two ground runners for enclosure protection. Two tracks on the top of the enclosure fit the ground runners in stacking deployment.



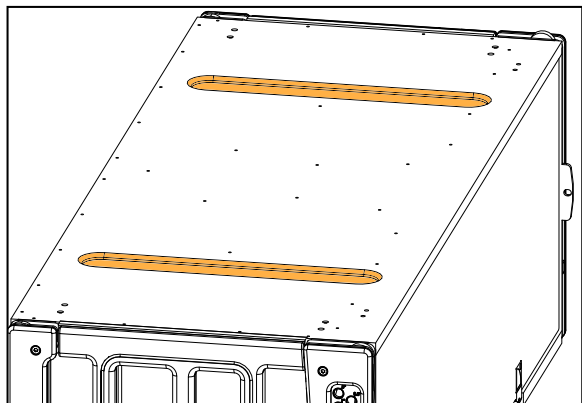
K1



K1-SB



K1



K1-SB

Rigging elements

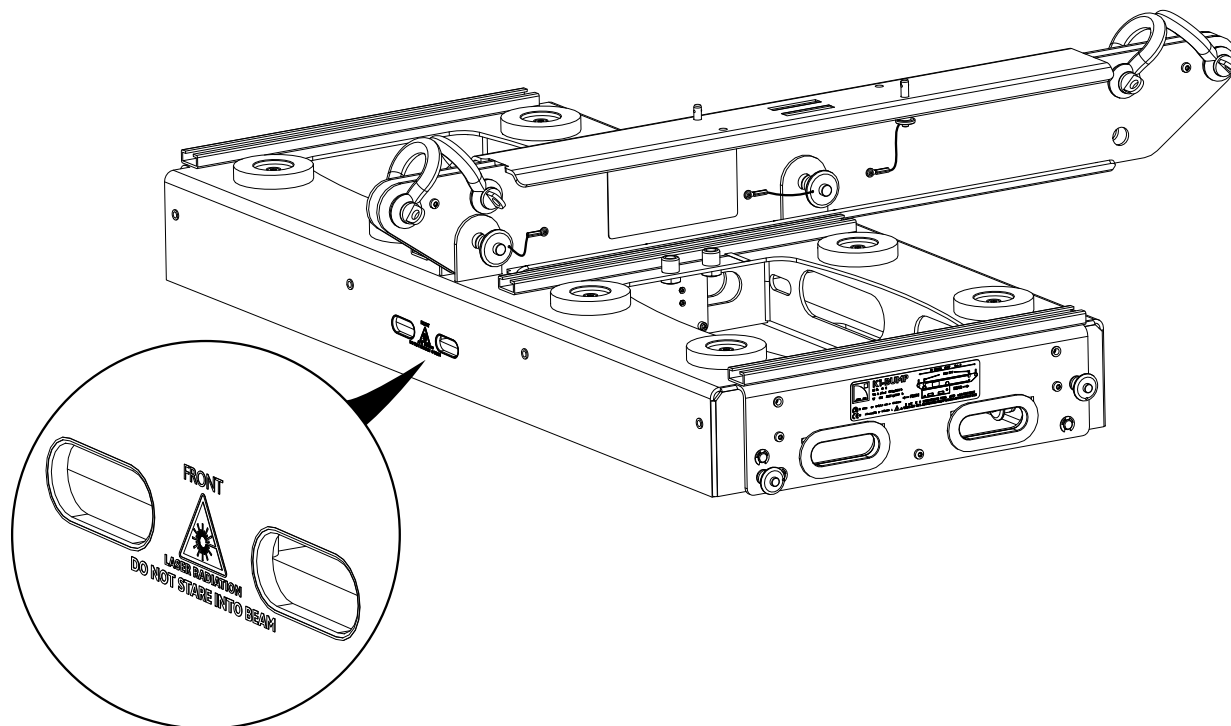
K1-BUMP

K1-BUMP is a rigging frame designed for flying K1/K1-SB enclosure arrays. It features a central rigging bar with four Ø 22 mm shackles to secure security slings, K1-BPCHAIN and K1-DELTA.

K1-BUMP also features a support plate to accommodate a laser inclinometer. See [K1-BUMP](#) (p.140).

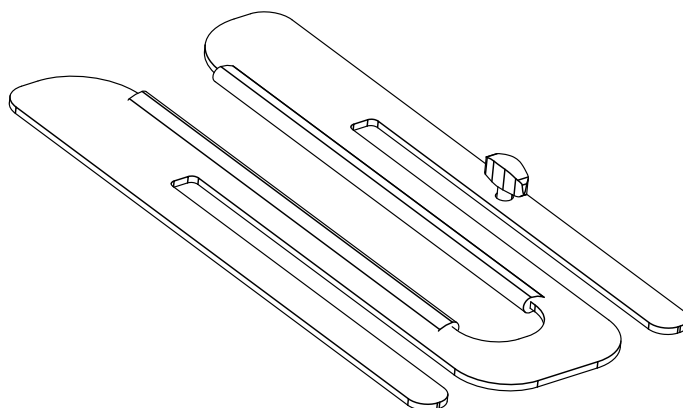


An identification sticker is present at the front of K1-BUMP to identify the front and the rear sides.

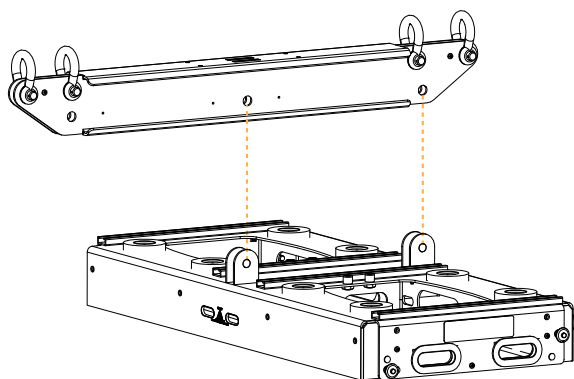


A rack stabilizer is available to mount LA-RAK II AVB touring racks.

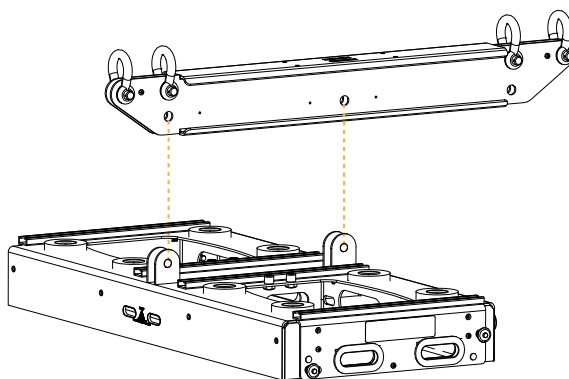
Refer to [Preparing LA-RAK II AVB touring racks](#).



The central bar of K1-BUMP has two holes each for site angle adjustments. Adjust the central bar position for a front side or a rear overhang of the enclosure array.

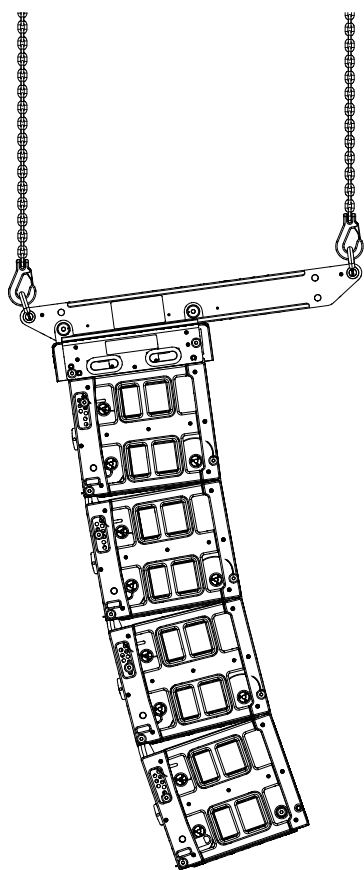


Front overhang

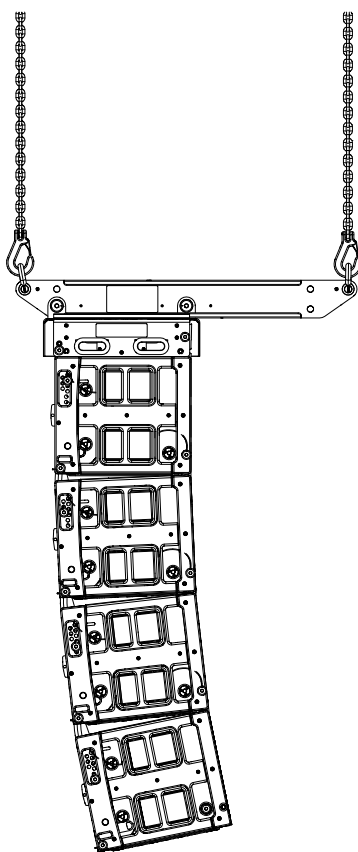


Rear overhang

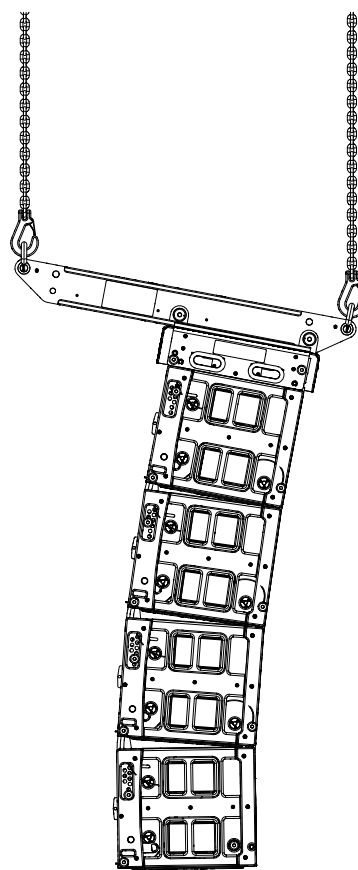
The array site angle depends on the relative heights between the K1-BUMP's front and rear hang points (motor chain lengths) and central bar configuration (front or rear overhang).



Rear overhang ($F > R$)

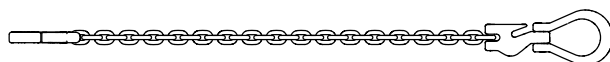


Rear overhang ($F = R$)

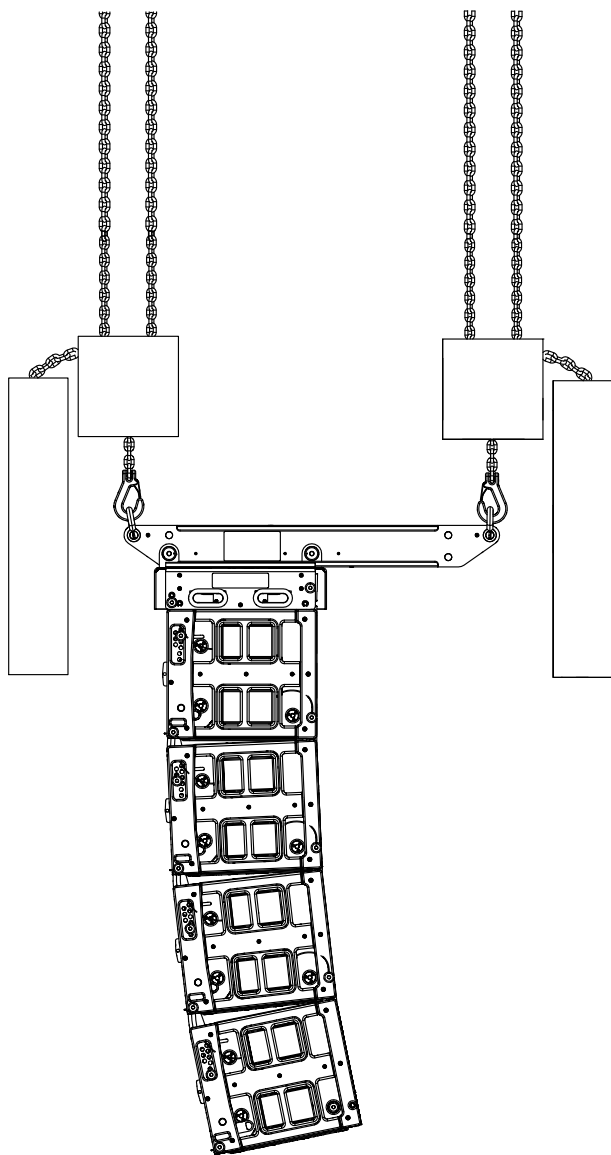


Front overhang ($F < R$)

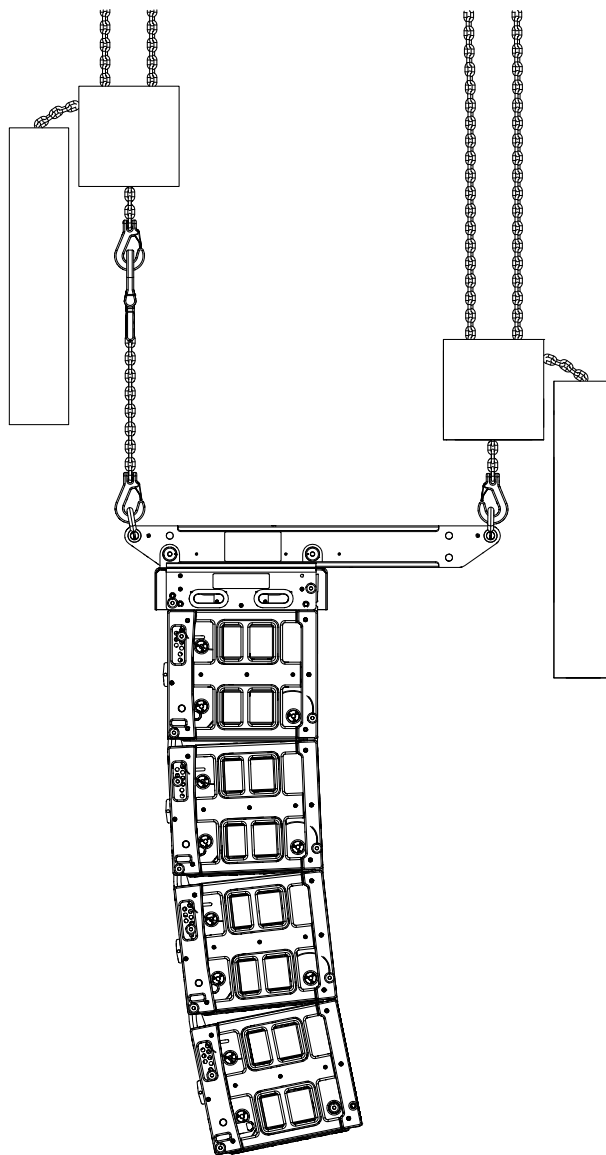
K1-BPCHAIN is an adjustable extension chain compatible with K1-BUMP or K2-BUMP.



K1-BPCHAIN has a WLL (Working Load Limit) of 3.15 t and is necessary for climbing-hoist setup. It allows to prevent the chain bag from lying in front of the enclosures located on top of the array.



Without K1-BPCHAIN



With K1-BPCHAIN

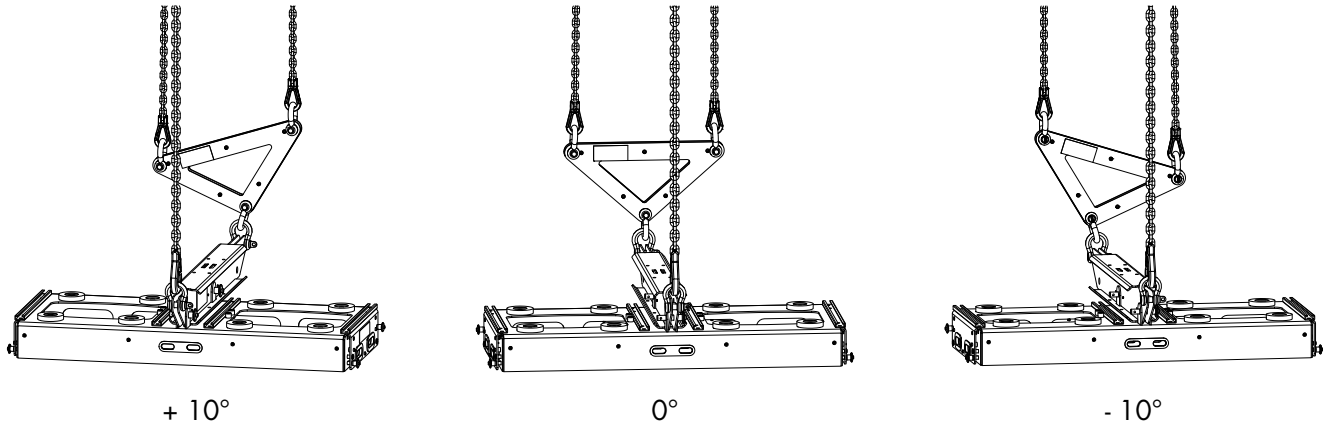
K1-DELTA

K1-DELTA is designed for azimuth angle setting.

Combined with K1-BUMP, K1-DELTA can be used to control the azimuth angle while K1-BUMP rigging bar controls the site angle.

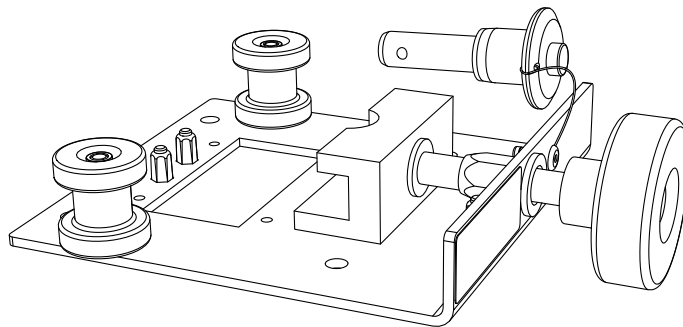
The recommended space between the two lifting points is 1 m / 33 ft.

By adjusting the height of both pickup points, it is possible to adjust the azimuth angle from -10° to $+10^{\circ}$.

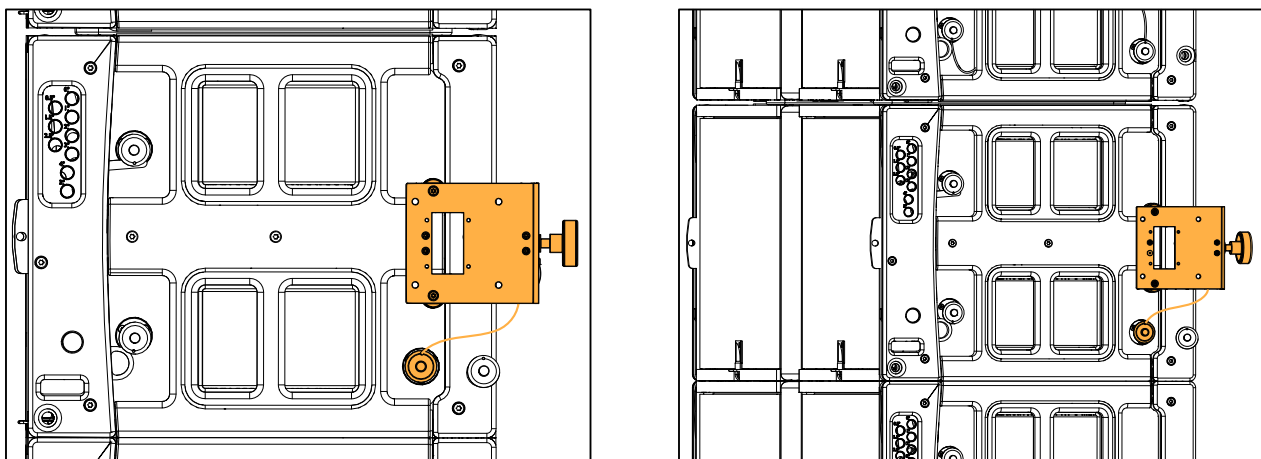


K1-LASERMOUNT

K1-LASERMOUNT is a support plate for mounting a laser inclinometer to the side of a K1 or K1-SB enclosure.



K1-LASERMOUNT is installed at the rear of the enclosure, on the right side.

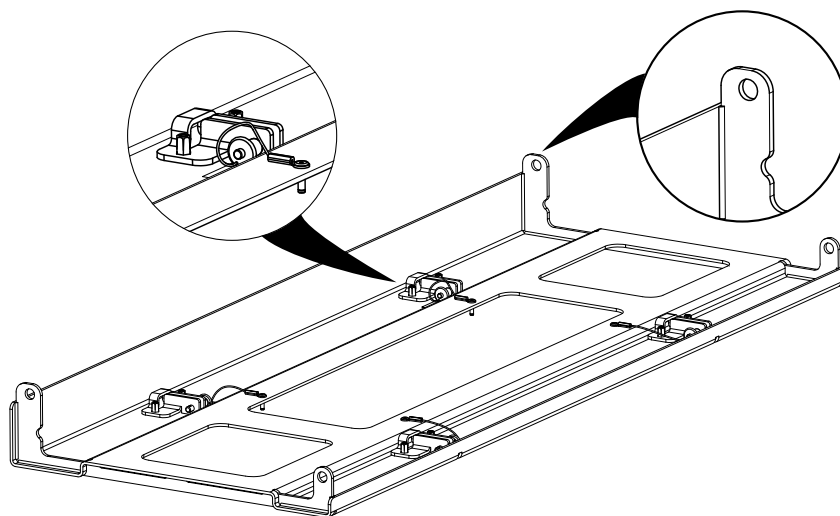


Refer to [Installing a laser inclinometer](#) (p.139).

KARA-DOWNK1

KARA-DOWNK1 is a rigging interface for Kara II under a K1 array. Refer to [Rigging a Kara II downfill array to the K1 array](#).

The four tabs on the top are compatible with the K1 rigging system. The four slits fitted with ball-locking pins accommodate the Kara II rigging arms.



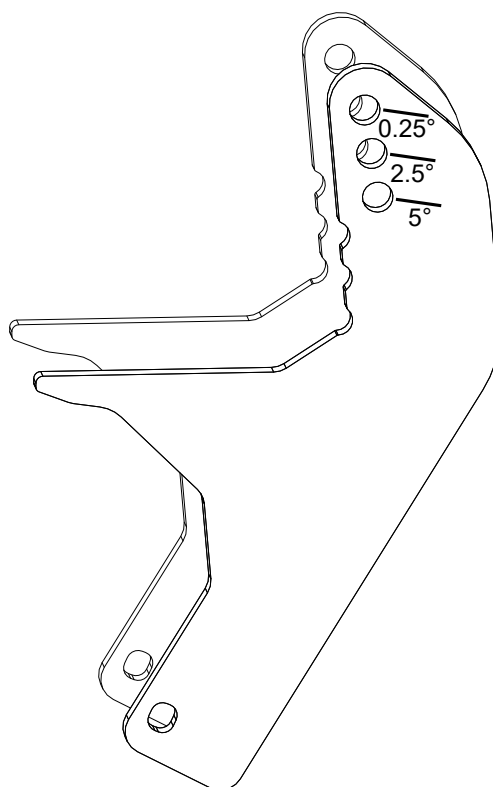
K2-LINK

K2-LINK is designed as an interface between the K1 and K2 rigging systems.

K2-LINK is used to hang K2 under K1, K1-SB, or K1-BUMP. Three holes at the top provide three angle settings between the top K2 enclosure and the bottom K1 element: 0.25°, 2.5°, and 5°.



To align the K2 and K1 element front faces, use the 0.25° hole.

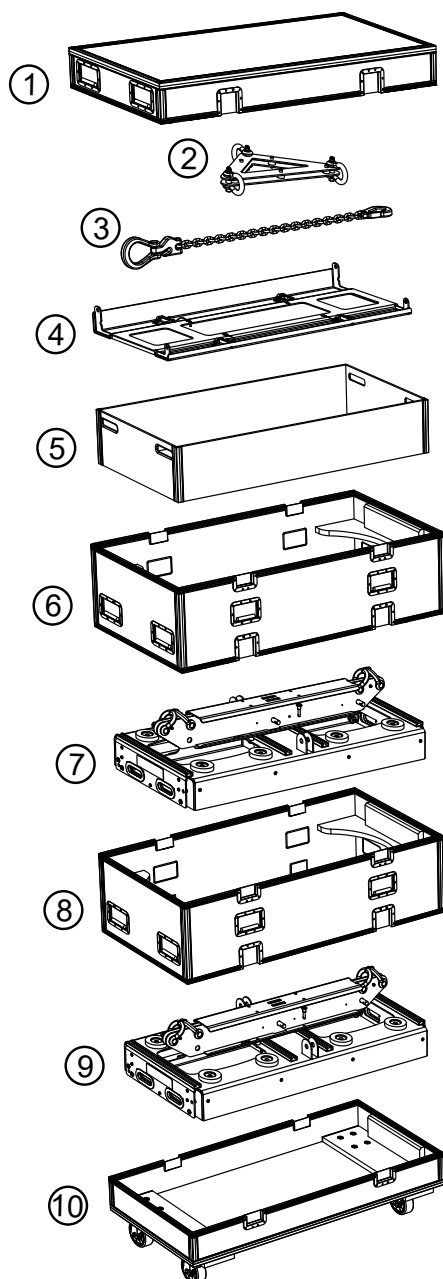


Transportation

K-BUMPFLIGHT

K-BUMPFLIGHT is a modular flight-case used to transport the accessories of the K1 system. It is designed to hook and lift K1-BUMP directly from the flight-case without any physical handling.

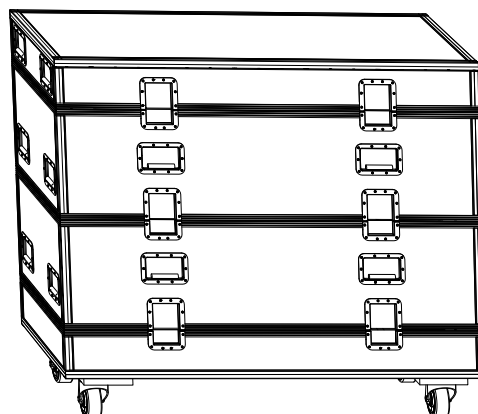
K-BUMPFLIGHT allows to transport up to two K1-BUMP and additional accessories in the removable tray:



1. K-BUMPFLIGHT cover
2. K1-DELTA
3. K1-BPCHAIN
4. KARA-DOWNK1
5. Removable tray
6. Compartment
7. K1-BUMP
8. Compartment
9. K1-BUMP
10. Wheeled bottom tray

K-BUMPFLIGHT can also be fitted with two additional compartments. In that case, it can transport up to four K1-BUMP and additional accessories in the removable tray.

The total height of the flight-case with two additional compartments is equivalent to the height of an array of four K1 or K1-SB enclosures placed on a K1-CHARIOT.



K1-CHARIOT

K1-CHARIOT is a chariot designed for the transportation and storage of blocks of up to four K1 or K1-SB.

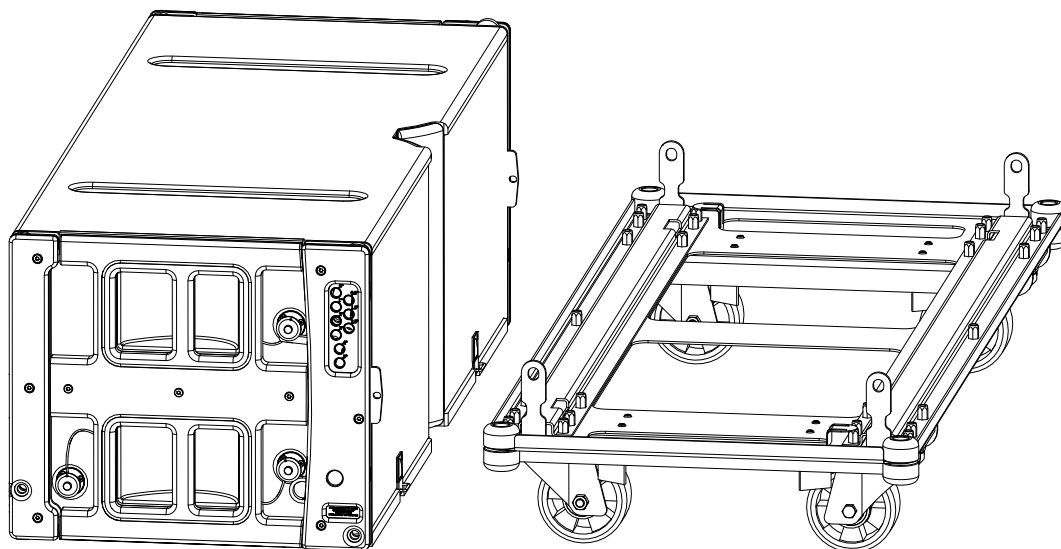


Do not move or transport more than four K1 or K1-SB on K1-CHARIOT.



K1-CHARIOT is intended for use only with L-Acoustics K1 and K1-SB

Use with other equipment may result in instability causing injury.



It features four rigging arms to secure the lower K1 or K1-SB.

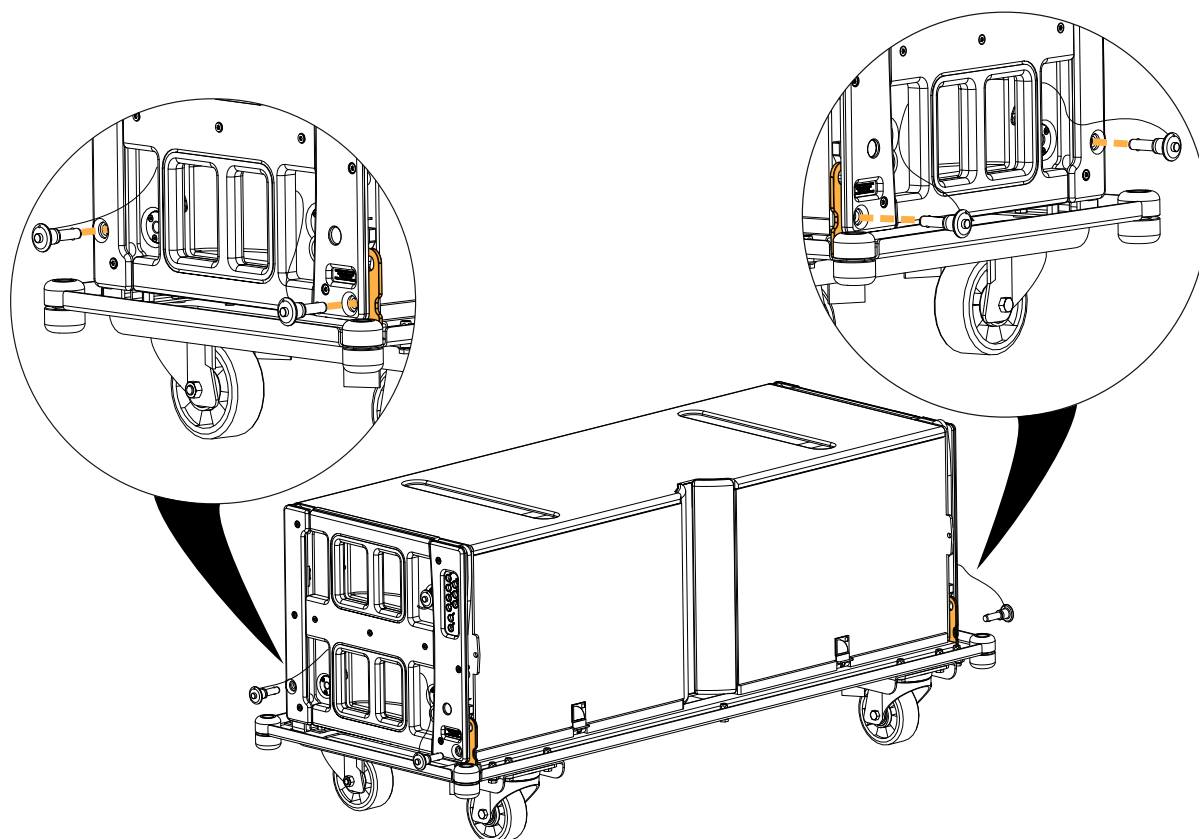


Enclosure can face either way.



Risk of severe injury

Remove all enclosures before performing any maintenance operation on K1-CHARIOT.



K1-PLA

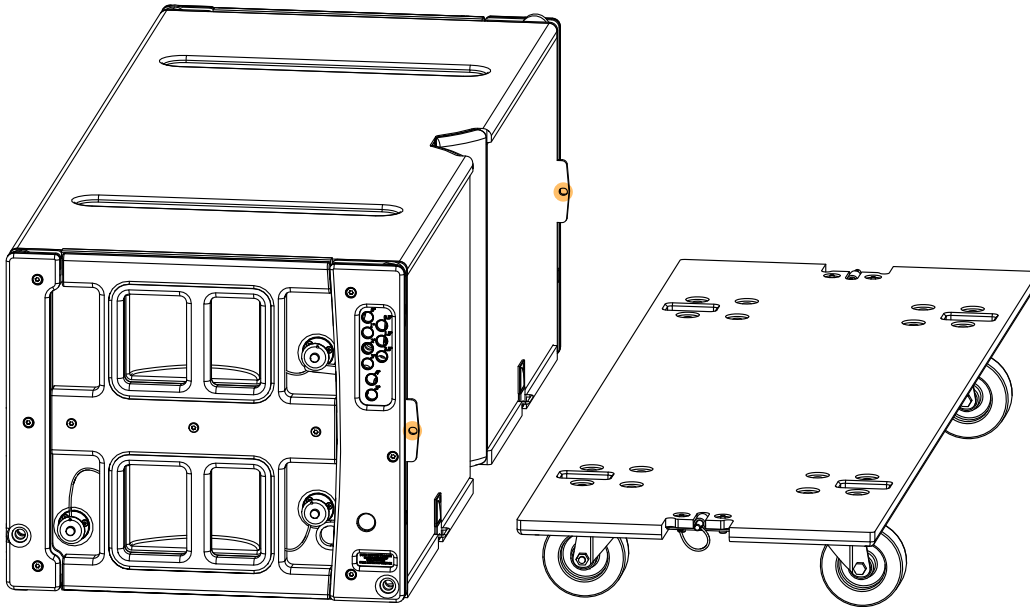
K1-PLA is a dolly board that can be secured to the front of a K1 or K1-SB with two ball-locking pins.



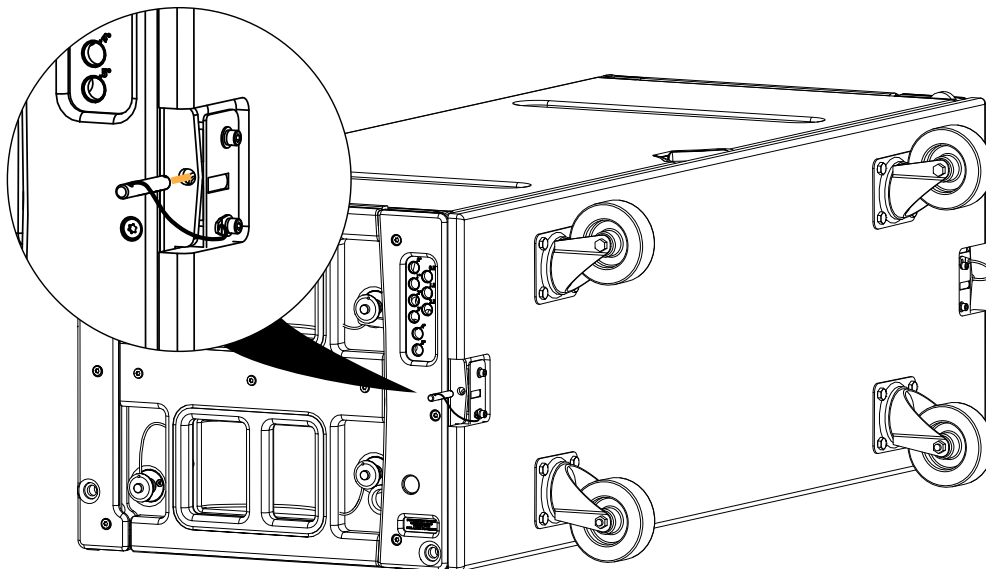
K1-PLA is intended for use only with L-Acoustics K1 or K1-SB.

Use with other equipment may result in instability causing injury.

Two holes, at the front of K1 or K1-SB, are available to secure a K1-PLA.



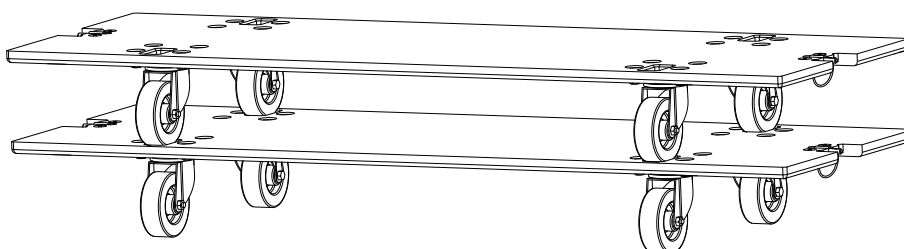
Use both K1-PLA ball-locking pins to secure K1.



Multiple K1-PLA can be stacked for storage.



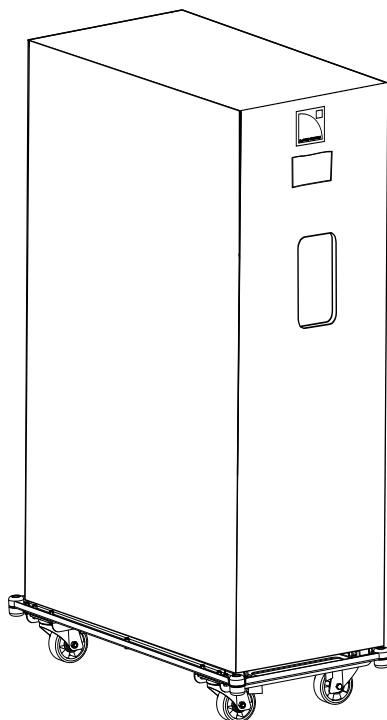
Turn the wheels outwards to stack K1-PLA.



K1-CHARIOTCOV

K1-CHARIOTCOV is a protective cover for a stack of four K1 or K1-SB enclosures.

A zipper at the rear provides access to the connector plate of each enclosure.

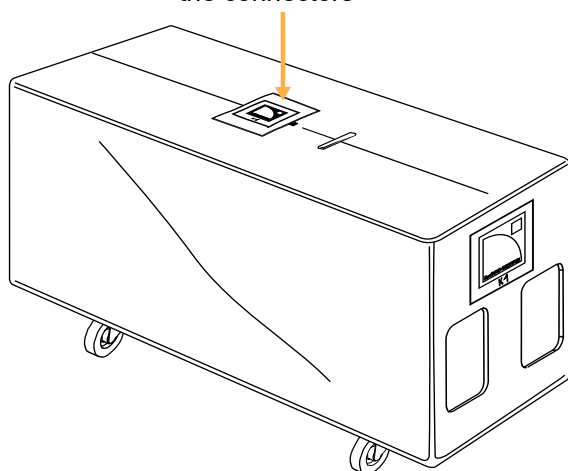


K1-COV/K1-SBCOV

K1-COV and K1-SBCOV are protective covers for one K1 or K1-SB enclosure on K1-PLA.

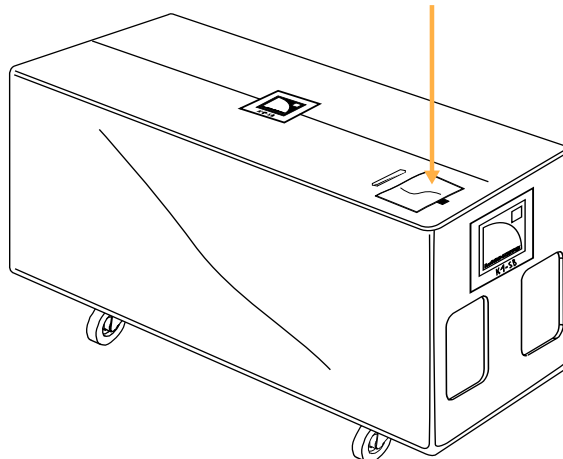
On each cover, a foldable flap provides access to the connector plate.

Foldable flap to access
the connectors



K1-COV

Foldable flap to access
the connectors



K1-SBCOV

Mechanical safety

Flown configurations

The K1 rigging system complies with 2006/42/EC: Machinery Directive. It has been designed following the guidelines of BGV-C1.

2006/42/EC: Machinery Directive specifies a safety factor of 4 against the rupture. The flown deployments described in this manual achieve a safety factor of **4 or more**.

Refer to Soundvision for the safety factor of a specific deployment.

The **safe limit** gives the maximum number of elements for which the safety factor is compliant with the 2006/42/EC: Machinery Directive, within the use defined in this manual and regardless of the other deployment parameters (site angles, inter-element angles, etc.).

The **maximum limit** gives the maximum number of elements for which the safety factor can be compliant with the 2006/42/EC: Machinery Directive, when the other deployment parameters provide the best mechanical conditions.

For mixed arrays refer to your Soundvision model.



Always refer to Soundvision for the safety factor of a mixed array.

When flying a K1 array with a Kara II downfill, the mechanical safety of all system elements must be considered. The indicated maximum limit applies to the K1 only.



The site angle limits with LA-RAK II AVB on top of a K1 array are -20°/+20°.



Do not implement a pullback on a K1 array with LA-RAK II AVB on top.



Do not implement a pullback on a K1 array with a Kara II downfill.

K1

configuration	rigging accessory	safe limit	maximum limit
flown	K1-BUMP	16	24
	K1-BUMP	14 K1 + 3 LA-RAK II AVB	23 K1 + 4 LA-RAK II AVB
flown with a downfill	KARA-DOWNK1	6 Kara II	

K1-SB

configuration	rigging accessory	safe limit	maximum limit
flown	K1-BUMP	20	24
	K1-BUMP	18 K1-SB + 3 LA-RAK II AVB	24 K1-SB + 4 LA-RAK II AVB

KS28

configuration	rigging accessory	maximum / safe limit
flown	KS28-BUMP	16

Other configurations

For other configurations, respect the recommended maximum limit for optimal stability.

K1

configuration	rigging accessory	safe limit	maximum limit
stacked	K1-CHARIOT	4	4

KS28

configuration	rigging accessory	maximum limit
ground-stacked	no rigging accessory or KS28-OUTRIG	4
stacked upright	no rigging accessory	2
stacked on chariot	KS28-CHARIOT	4

Assessing mechanical safety



Mechanical safety of the rigging system

Before any installation, always model the system in Soundvision and check the **Mechanical Data** section for any stress warning or stability warning.

In order to assess the actual safety of any array configuration before implementation, refer to the following warnings:



Rated working load limit (WLL) is not enough

The rated WLL is an indication of the element resistance to tensile stress. For complex mechanical systems such as loudspeaker arrays, WLLs cannot be used per se to determine the maximum number of enclosures within an array or to assess the safety of a specific array configuration.

Mechanical modeling with Soundvision

The working load applied to each linking point, along with the corresponding safety factor, will depend on numerous variables linked to the composition of the array (type and number of enclosures, splay angles) and the implementation of the flying or stacking structure (number and location of flying points, site angle). This cannot be determined without the complex mechanical modeling and calculation offered by Soundvision.

Assessing the safety with Soundvision

The overall safety factor of a specific mechanical configuration always corresponds to the lowest safety factor among all the linking points. Always model the system configuration with the Soundvision software and check the **Mechanical Data** section to identify the weakest link and its corresponding working load. By default, a stress warning will appear when the mechanical safety goes beyond the recommended safety level.

Safety of ground-stacked arrays in Soundvision

For ground-stacked arrays, a distinct stability warning is implemented in Soundvision. It indicates a tipping hazard when the array is not secured to the ground, stage or platform. It is the user's responsibility to secure the array and to ignore the warning.

Additional safety for flown arrays

When flying an array, use available holes to implement a secondary safety.

Considerations must be given to unusual conditions

Soundvision calculations are based on usual environmental conditions. A higher safety factor is recommended with factors such as extreme high or low temperatures, strong wind, prolonged exposition to salt water, etc. Always consult a rigging specialist to adopt safety practices adapted to such a situation.

Loudspeaker configurations

Line source

Deployed as a line source, the system operates over the nominal bandwidth of the K1 enclosure.

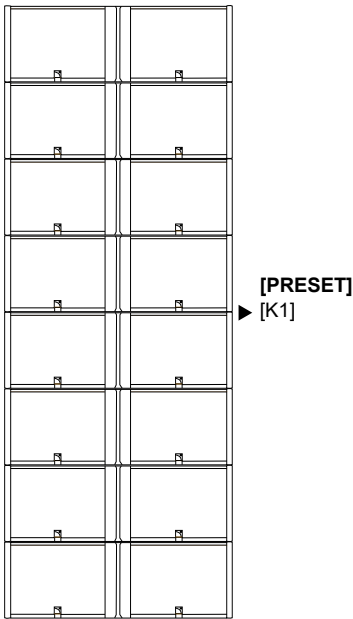
Two configurations are possible:

- K1 line source
- K1/K1-SB line source: enhanced LF throw

By providing the K1-SB with the same frequency response as the K1 low section, the [K1SB_X] preset allows the K1-SB enclosure to be used as an LF line source element, increasing the length of the sub-low line source.

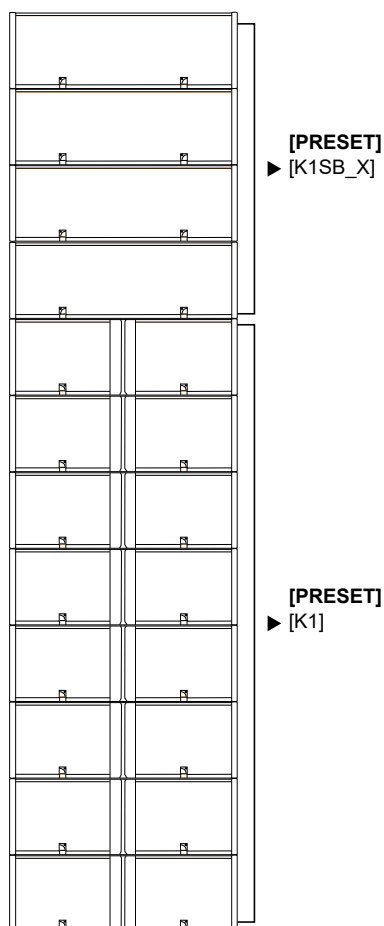
The K1 and K1-SB enclosures are driven by the LA12X amplified controllers.

Standalone K1 line source



Enclosure	K1
Preset	[K1]
Frequency range (-10 dB)	35 Hz - 20 kHz

K1/K1-SB line source



Enclosure	K1	K1-SB
Preset	[K1]	[K1SB_X]
Frequency range (-10 dB)	35 Hz - 20 kHz	
Recommended ratio	2 K1 : 1 K1-SB	
Minimum line length	8 K1 + 4 K1-SB	

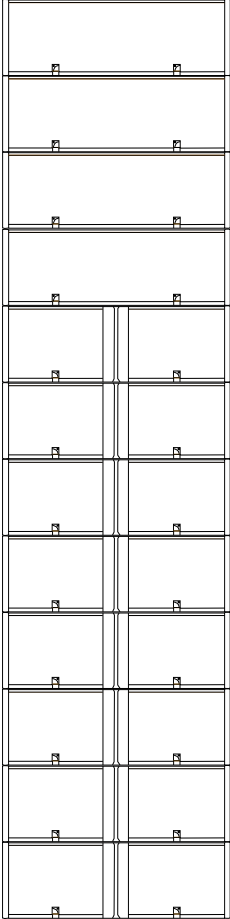
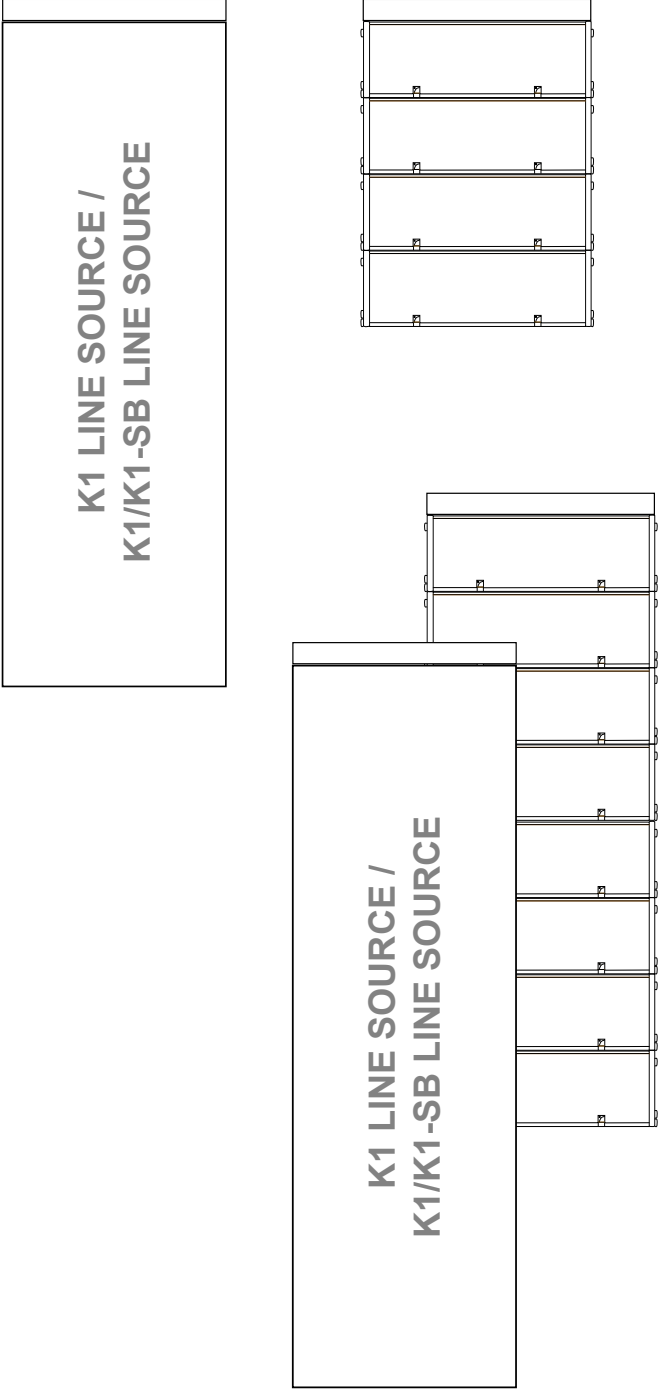
! When using [K1] with [K1SB_X], do not add any delay value between the K1 and K1-SB elements of a same line source.

K1-SB applications

There are two distinct applications for K1-SB in a K1 system:

- As an LF extension in a line source for enhanced throw, using the [K1SB_X] preset with K1.
- As a subwoofer for increased impact, using the [K1SB_60] preset.

i Both applications of K1-SB can be combined in the same configuration.

K1-SB as an LF extension in a line source	K1-SB as a subwoofer
	
ENHANCED THROW	INCREASED IMPACT

Additional subwoofer system

A K1 line source or a K1/K1-SB line source can be deployed with additional subwoofer enclosures to provide increased sub-low resources to demanding applications.

Two subwoofer systems are available:

- K1-SB for increased impact
- KS28 for infra extension

The recommended ratio is one K1 for one subwoofer, whether using K1-SB subwoofers only, KS28 subwoofers only, or a combination of both. The [K1SB_60] and [KS28_60] presets provide the subwoofers with an upper frequency limit at 60 Hz for an optimal frequency coupling with the line source.

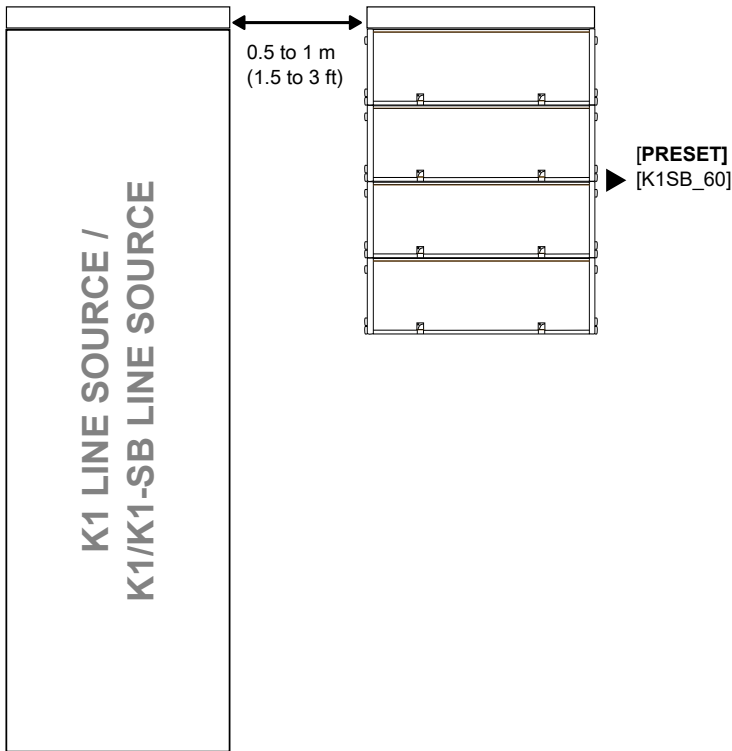
K1-SB and KS28 are driven by the LA12X amplified controller.

K1-SB

K1-SB provides an extension of the bandwidth in the low end, down to 30 Hz. Depending on the deployment, LF rejection can be produced. Two deployments are available in this configuration:

- K1-SB beside the K1-SB or K1/K1-SB line source: side LF rejection (polarized)
- K1-SB behind the K1 or K1/K1-SB line source: rear LF rejection (cardioid)

Line source with K1-SB beside



Enclosure	K1	K1-SB in line source	K1-SB as subwoofer
Preset	[K1]	[K1SB_X]	[K1SB_60]
Frequency range (-10 dB)	30 Hz - 20 kHz		
Recommended ratio	1 K1 : 1 subwoofer		
Recommended distance between front panels	Between 0.5 m (1.5 ft) and 1 m (3 ft)		

- **Delay values**

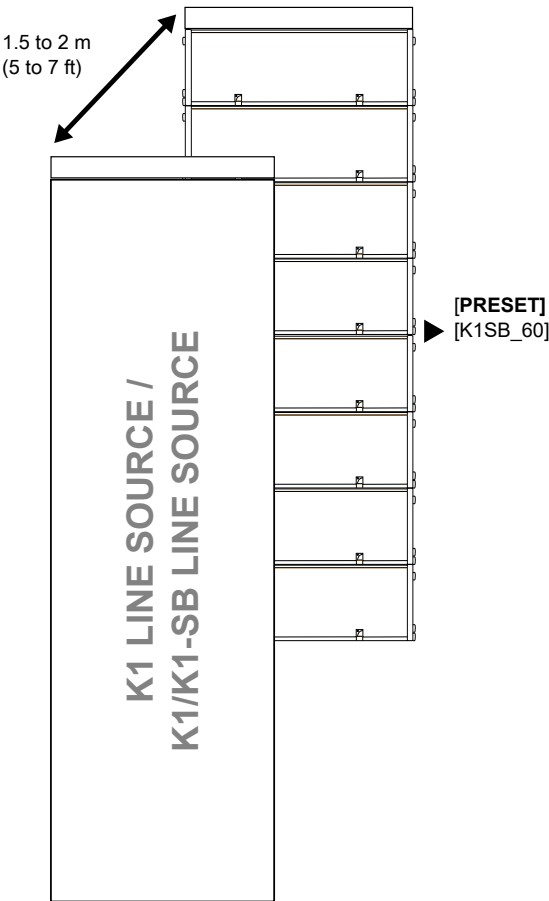
Do not forget to add the pre-alignment and geometric delays depending on the configuration.
- 

When using [K1] with [K1SB_X], do not add any delay value between the K1 and K1-SB elements of a same line source.

Pre-alignment delays

presets	pre-alignment delay values and polarity settings	
[K1] + [K1SB_60]	K1 = 6 ms 	K1-SB = 0 ms 

Line source with K1-SB behind



Enclosure	K1	K1-SB in line source	K1-SB as subwoofer
Preset	[K1]	[K1SB_X]	[K1SB_60]
Frequency range (-10 dB)	30 Hz - 20 kHz		
Recommended ratio	1 K1 : 1 subwoofer		
Recommended distance between front panels	Between 1.5 m (5 ft) and 2 m (7 ft)		

- **Delay values**

Do not forget to add the pre-alignment and geometric delays depending on the configuration.
- 

When using [K1] with [K1SB_X], do not add any delay value between the K1 and K1-SB elements of a same line source.

Pre-alignment delays

K1 + K1-SB

presets	pre-alignment delay values and polarity settings	
[K1] + [K1SB_60]	K1 = 6 ms 	K1-SB = 0 ms 

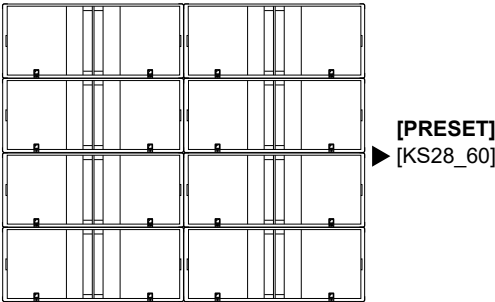
Geometric delays

1.5 m (5 ft)	Line source = 4.5 ms
2 m (7 ft)	Line source = 6 ms

KS28

The KS28 provides an extension of the bandwidth in the low end, down to 25 Hz.

Line source with KS28



Enclosure	K1	K1-SB in line source	KS28
Preset	[K1]	[K1SB_X]	[KS28_60]
Frequency range (-10 dB)	25 Hz - 20 kHz		
Recommended ratio	1 K1 : 1 subwoofer		

- 

Grouping subwoofers
Place the subwoofer enclosures side by side. If not possible, the maximum distance between two adjacent acoustic centers must be 2.8 m (9.2 ft) or 1.7 m (5.6 ft) if the upper frequency limit of the subwoofer system is at 60 Hz or 100 Hz, respectively.
- 

Use [xxxx_xx_C] or [xxxx_xx_Cx] on a reversed subwoofer in a cardioid configuration
The cardioid configuration consists in reversing 1 element in an array of 4 subwoofers.
Refer to the subwoofer owner's manual and to the **Cardioid configurations** technical bulletin.
- 

Delay values
Do not forget to add the pre-alignment and geometric delays depending on the configuration.
- 

When using [K1] with [K1SB_X], do not add any delay value between the K1 and K1-SB elements of a same line source.

Pre-alignment delays**K1 + KS28**

presets	pre-alignment delay values and polarity settings			
[K1] + [KS28_60]	K1 = 0.5 ms		KS28 = 0 ms	
[K1] + [KS28_60_C]	K1 = 6 ms		KS28 = 0 ms	
[K1] + [KS28_60_Cx]	K1 = 4 ms		KS28 = 0 ms	

K1 + K1-SB + KS28

presets	pre-alignment delay values and polarity settings				
[K1] + [K1SB_X] + [KS28_60]	K1 = 0 ms		K1-SB = 0 ms		KS28 = 0 ms 
[K1] + [K1SB_X] + [KS28_60_C]	K1 = 5.5 ms		K1-SB = 5.5 ms		KS28 = 0 ms 
[K1] + [K1SB_X] + [KS28_60_Cx]	K1 = 3.5 ms		K1-SB = 3.5 ms		KS28 = 0 ms 

Additional downfill element

All K1 system configurations can be combined with an additional Kara II or K2 line source downfill system. This allows an extension of the vertical coverage to the closer audience.

Kara II

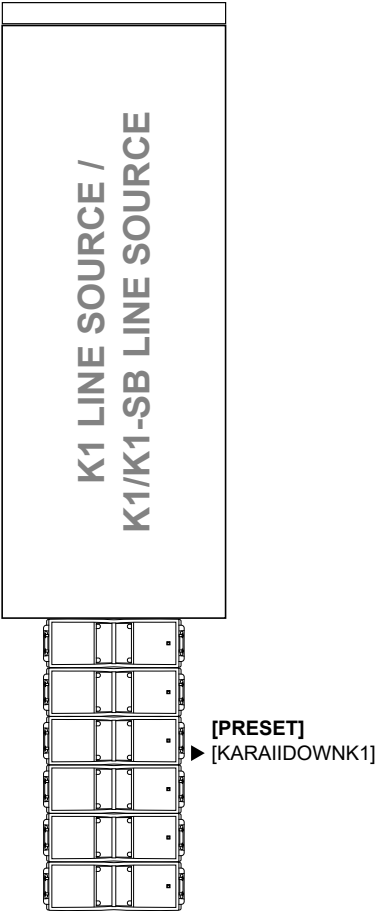
The [KARAIIDOWNK1], [KARAIIDOWNK1 70] and [KARAIIDOWNK1 90] presets feature a high-pass filter at 55 Hz for the low section, along with specific delay settings, in order to optimize the acoustic coupling between the Kara II and K1 line sources.

The Kara II enclosure is driven by the LA4X / LA12X amplified controllers.

! The [KARAIIDOWNK1] preset is optimized for a **110°** fins setting on Kara II.

i Fins settings for downfill
In most venues, the 110° fins settings must be used for downfills.
In case there is a raised area in front of the scene, such as a catwalk, the 70° or 90° settings can be used to divert SPL from this area.

Line source with Kara II downfill system



Enclosure	K1	K1-SB in line source	Kara II
Preset	[K1]	[K1SB_X]	[KARAIIDOWNK1] [KARAIIDOWNK1 70] [KARAIIDOWNK1 90]
Frequency range (-10 dB)	35 Hz - 20 kHz		
Kara II array	Up to 6 Kara II enclosures		

-  Do not add any delay between the K1 and Kara II elements of a mixed line source.
-  When using [K1] with [K1SB_X], do not add any delay value between the K1 and K1-SB elements of a same line source.
-  **Using the Kara II system**
Refer to the **Kara II owner's manual** for the operating modes of Kara II as a main system.

K2

The [K2 110] preset provides K2 with the same horizontal coverage as K1 for optimal downfill.

The K2 enclosure is driven by the LA4X / LA12X amplified controller.

i Fins settings for downfill

In most venues, the 110° fins settings must be used for downfills.

In case there is a raised area in front of the scene, such as a catwalk, the 70° or 90° settings can be used to divert SPL from this area.

Line source with K2 downfill system



Enclosure	K1	K1-SB in line source	K2
Preset	[K1]	[K1SB_60]	[K2 70][K2 90][K2 110]
Frequency range (-10 dB)	35 Hz - 20 kHz		

- !** Do not add any delay between the K1 and K2 elements of a mixed line source.
- !** When using [K1] with [K1SB_X], do not add any delay value between the K1 and K1-SB elements of a same line source.
- i Using the K2 system**
Refer to the **K2 owner's manual** for the operating modes of K2 as a main system.

Inspection and preventive maintenance

How to do preventive maintenance

Inspect the system before any deployment and after any corrective maintenance operation.

Perform preventive maintenance at least once a year.

Rigging and hardware

Perform the [Rigging part inspection](#) (p.42) on each rigging part.

Use the [Mechanical system overview](#) (p.43) to identify critical parts of the system and apply the specific checks described in the [Inspection references](#) (p.49).

Do the [Rigging check](#) (p.52).

If any parts are damaged, contact your L-Acoustics representative for further instructions.

Acoustics

Perform the [Enclosure check](#) (p.58).

Perform the [Listening test](#) (p.60) to detect any degradation in sound quality.

If necessary, refer to the [Corrective maintenance](#) (p.106) section for speaker repair kits and maintenance instructions.

Rigging part inspection

About this task

For critical rigging parts, use the [Inspection references](#) (p.49) for comparison and specific manipulations.

The term "rigging part" comprises:

- lifting accessories such as clamps and shackles
- rigging accessories such as rigging frames, rigging interfaces, and brackets
- fasteners used for assembling two products together such as ball-locking pins, rigging axes, and safety pins
- rigging elements integrated in the product such as rigging arms and rails
- transportation accessories

This inspection procedure covers only L-Acoustics products. To inspect other products that are part of the lifting chain, refer to the manufacturer's instructions.

Prerequisite

Perform the inspection in a well-lit environment.

Procedure

1. Check that the rigging part is present.
2. If applicable, disassemble the rigging part from the enclosure or the rigging accessory.

Check that the tethers are intact and safely secured.

3. Inspect the part from every side.

Compare with the **reference pictures**.

Check for:

- corrosion
- wear and cracks
- bends and dents
- holes
- missing safety cues
- missing identification labels
- missing or loose fasteners



Replacing screws

If a screw is loose, remove and replace it.

Always use the new screws provided in the repair kit.

If no new screw is available, add blue threadlocker before reusing the screw.

Do not apply more than the indicated torque.

4. Check the **geometry** of the part to identify critical deformations.

Place the rigging part on a flat surface or hold a level against it.

5. Check the **moving parts**.

Make sure that the mechanism engages correctly.

What to do next

If a problem is detected, perform the authorized maintenance operations or contact your L-Acoustics representative.

Mechanical system overview

Critical parts of the lifting chains are highlighted.



The  indicates a visual inspection. The  indicates a functional check.



Perform the [Rigging part inspection](#) (p.42) on critical parts.

For each part, refer to the [Inspection references](#) (p.49).



Replacing screws

If a screw is loose, remove and replace it.

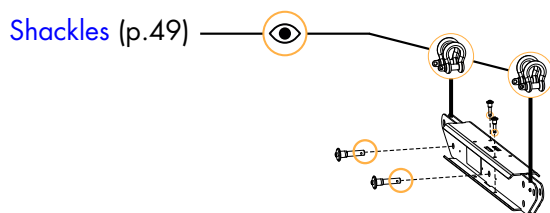
Always use the new screws provided in the repair kit.

If no new screw is available, add blue threadlocker before reusing the screw.

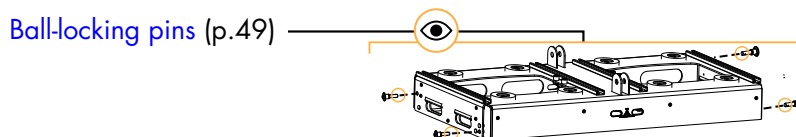
Do not apply more than the indicated torque.

K1 mixed array with K1-BUMP and K1-SB

Refer to [Inspection references](#) (p.49).

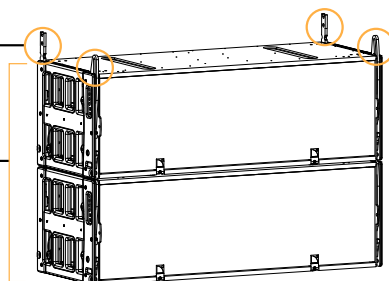


Ball-locking pins (p.49)



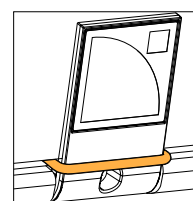
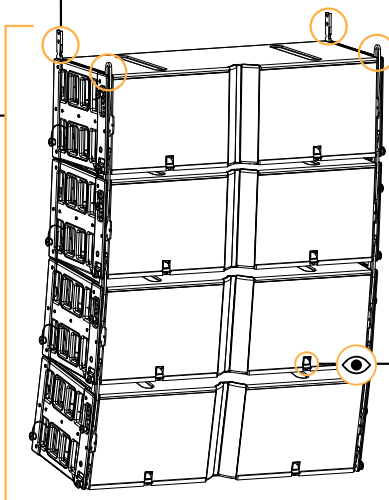
- Rear rigging arms rotate correctly
- Front rigging arms slide correctly

Ball-locking pins (p.49)



Rigging check (p.52)

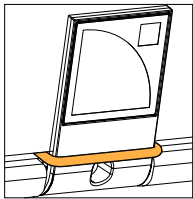
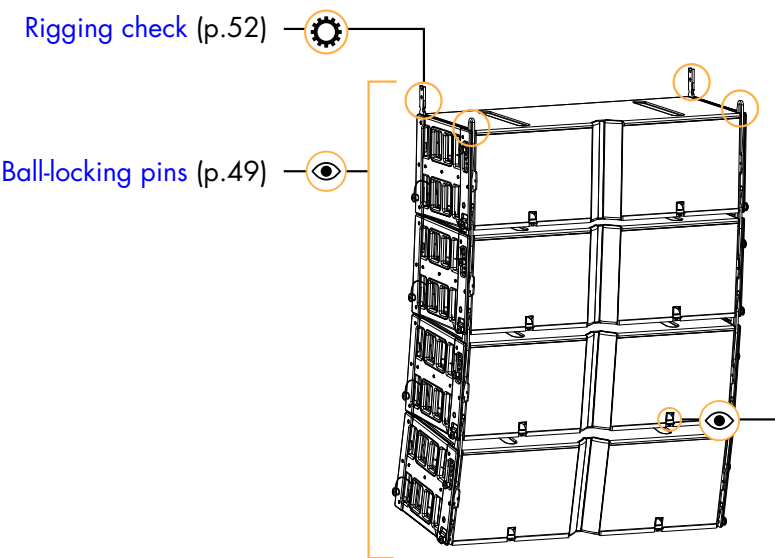
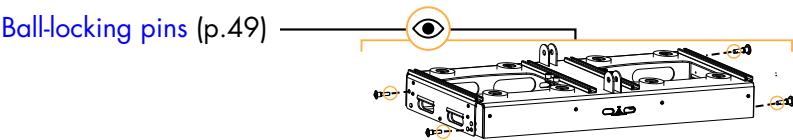
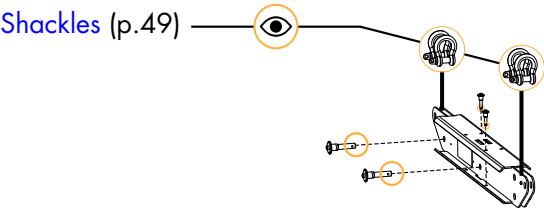
Ball-locking pins (p.49)



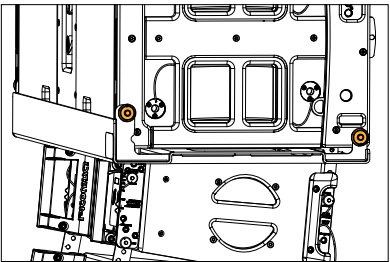
Grill safety

K1 array with K1-BUMP, KARA-DOWNK1 and Kara II

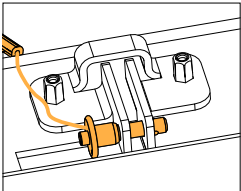
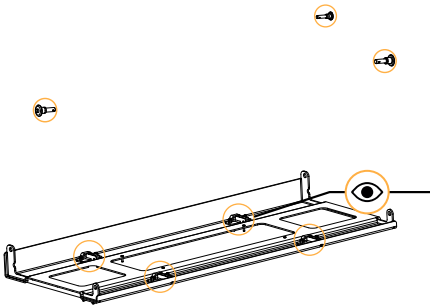
Refer to [Inspection references](#) (p.49).



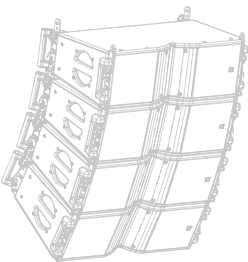
Grill safety



Ball-locking pins (p.49)



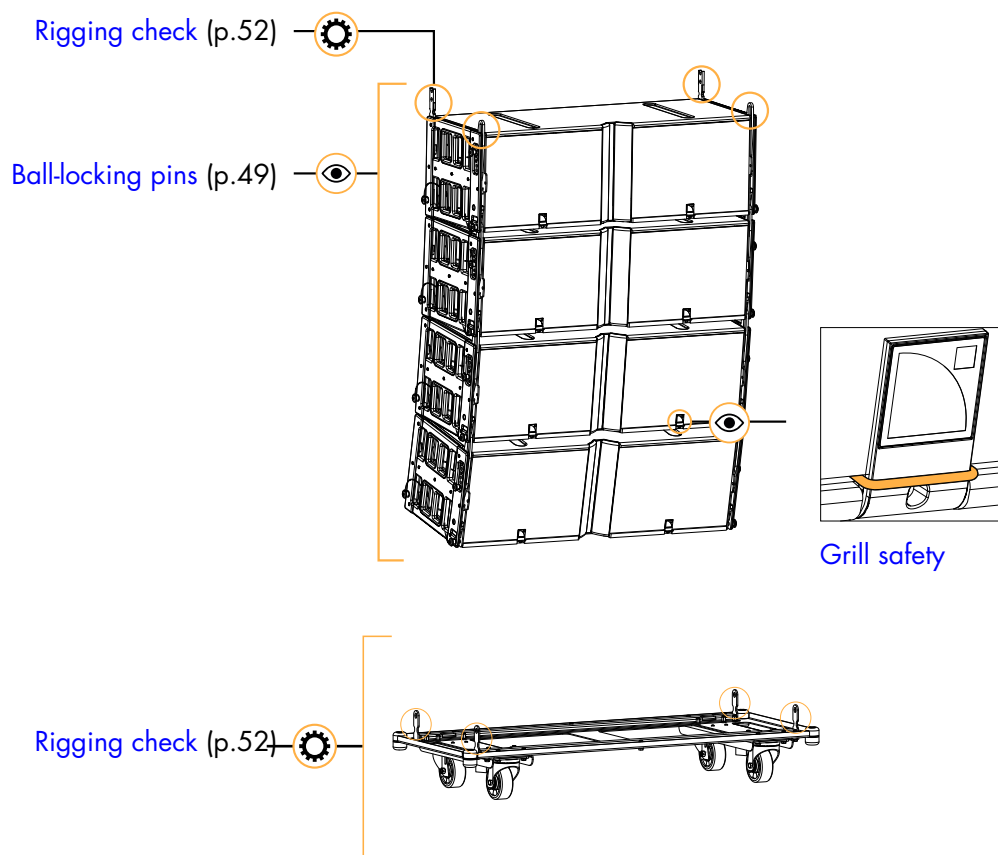
Ball-locking pins (p.49)



Refer to the **Kara II owner's manual** for more information.

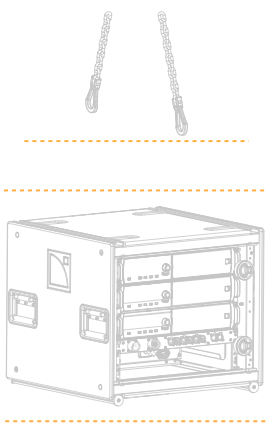
K1 array with K1-CHARIOT

Refer to [Inspection references](#) (p.49).



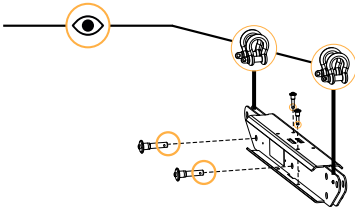
K1 array with K1-BUMP and LA-RAK II AVB

Refer to [Inspection references](#) (p.49).



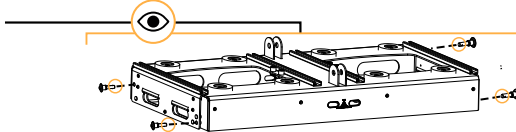
Refer to the **LA-RAK II AVB owner's manual** for more information.

Shackles (p.49)

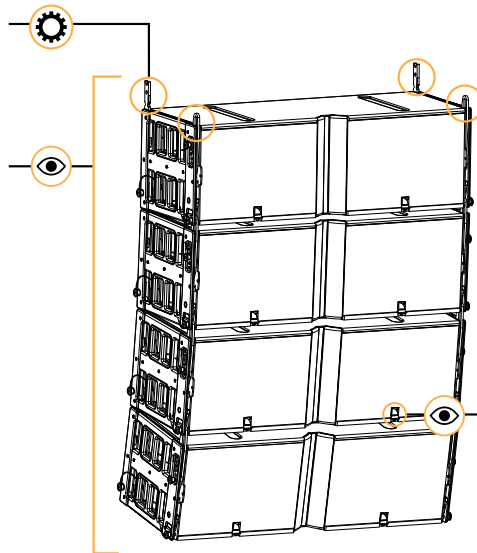


Ball-locking pins (p.49)

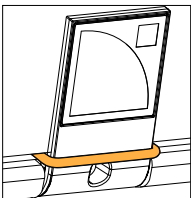
Ball-locking pins (p.49)



Rigging check (p.52)



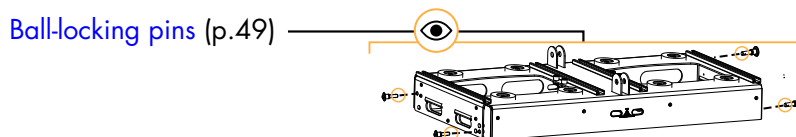
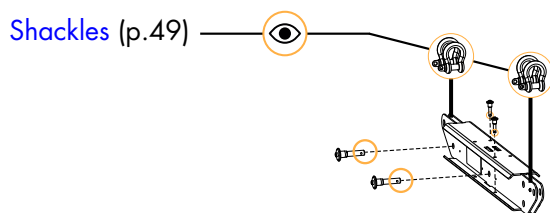
Ball-locking pins (p.49)



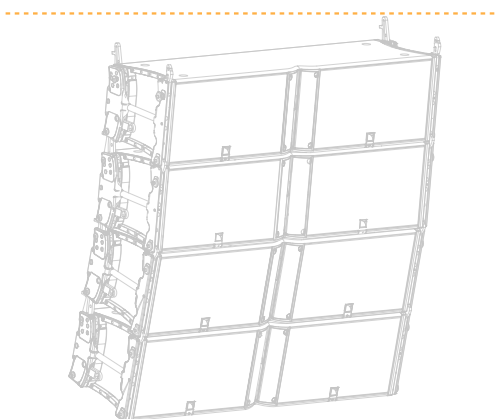
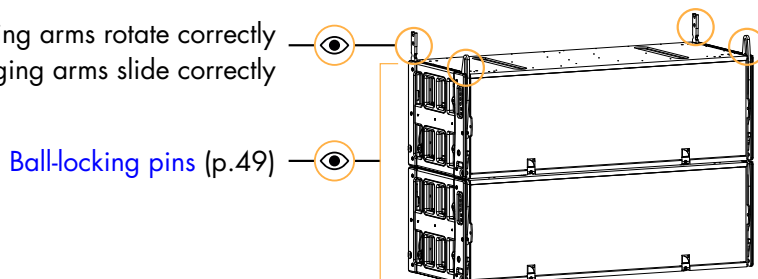
Grill safety

K2 mixed array with K1-BUMP, K1-SB, and K2-LINK

Refer to [Inspection references](#) (p.49).



- Rear rigging arms rotate correctly
- Front rigging arms slide correctly



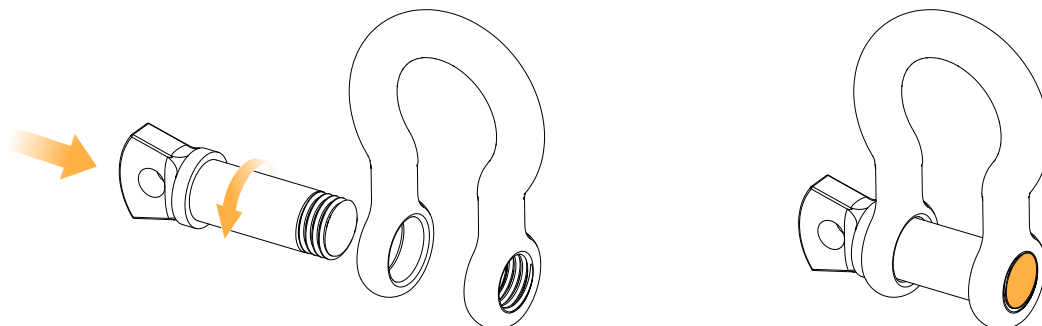
Refer to the **K2 owner's manual** for more information.

Inspection references

Shackles

Moving parts

Drive the shackle axis in its lodging. Make sure that the end is flush with the shackle.

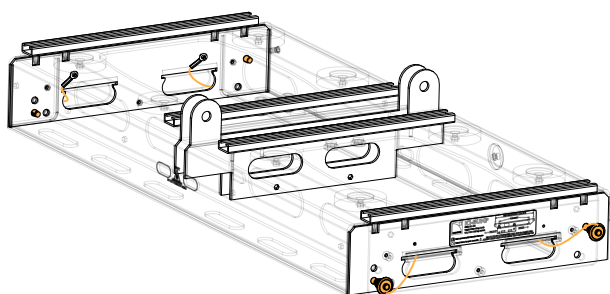


Related tasks

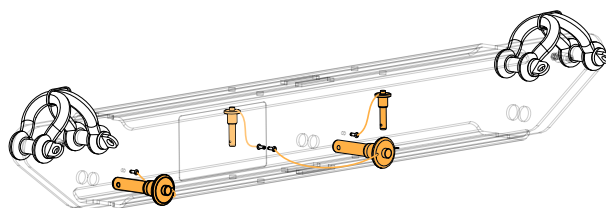
[Rigging part inspection](#) (p.42)

Ball-locking pins

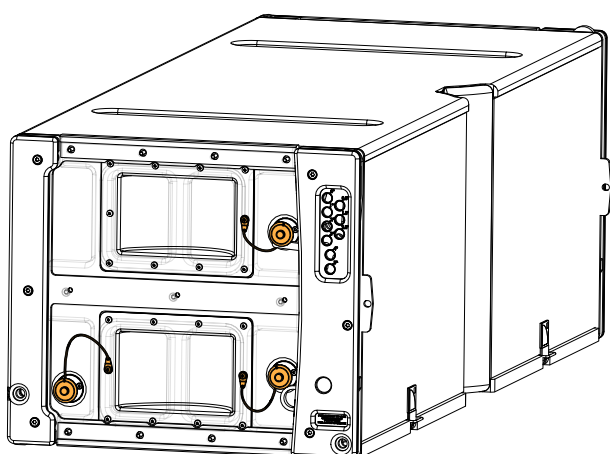
Reference pictures



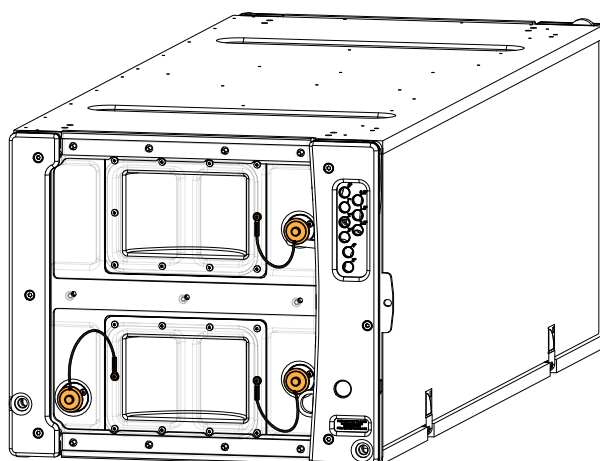
tether (K1-BUMP)



tether (K1-BUMP rigging bar)



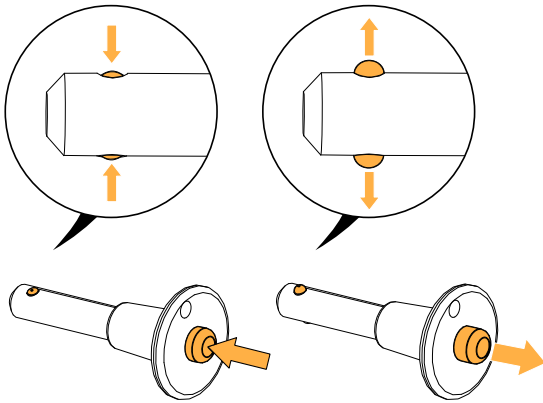
tether (K1)



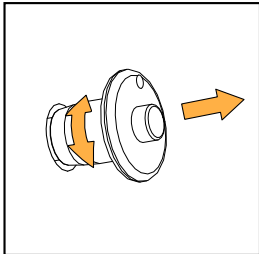
tether (K1-SB)

Moving parts

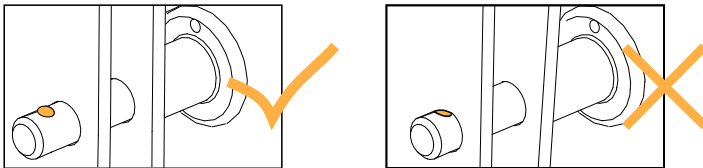
1. Press the push button. The ball-locking mechanism is retracted.
2. Release the button. The ball-locking mechanism is activated.



3. Insert the pin in each storage and rigging hole. Pull and rotate the pin. The pin must remain inside the hole.



If the pin is inserted in two plates, the ball must pass through both plates and lock the pin in place.



If the check fails, **immediately withdraw the product from use and contact L-Acoustics.**

Related tasks

[Rigging part inspection](#) (p.42)

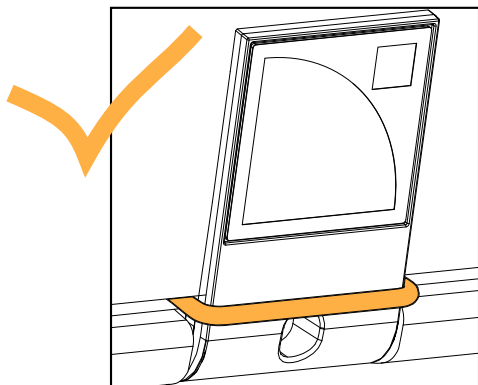
Grill safety

Visually and mechanically inspect the grill safety.

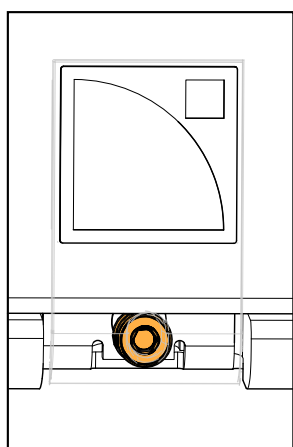
Moving parts

Procedure

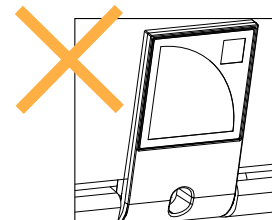
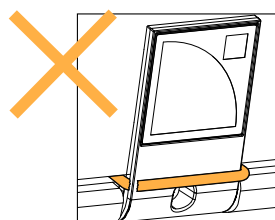
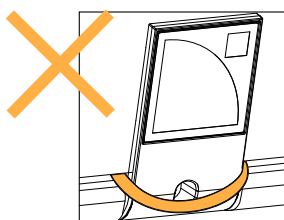
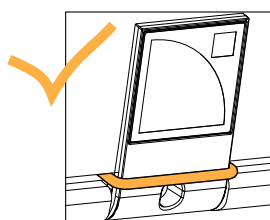
1. Make sure that the O-rings are not damaged.



2. Make sure that the screw is tightened.



3. Pull on the O-rings:
 - Make sure they do not break.
 - Make sure they are not loose.



If the O-rings are damaged or missing, refer to the **D/R - Grill (p.109)** to replace them.

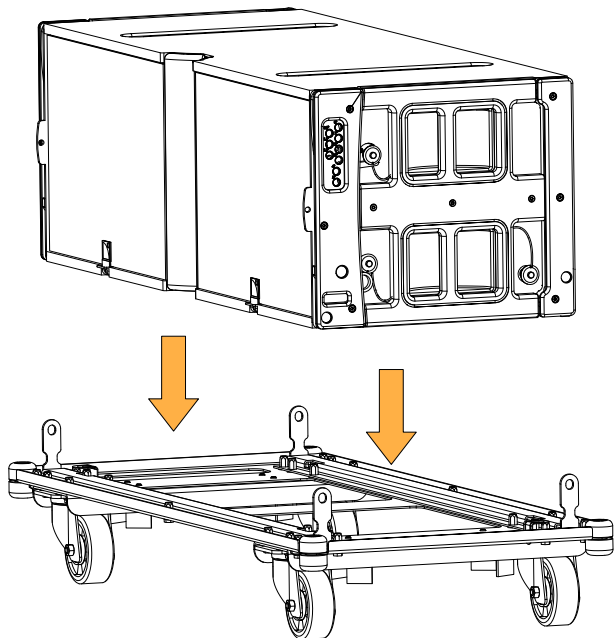
Rigging check

Procedure

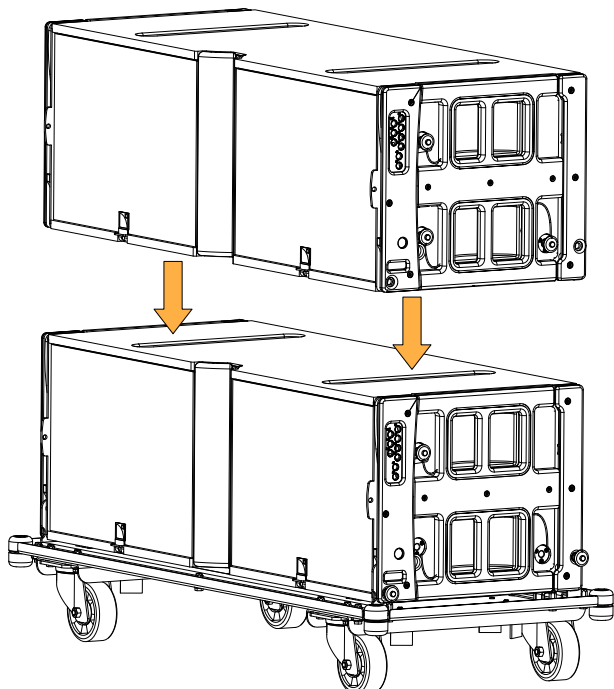
1. Position a K1 on K1-CHARIOT.



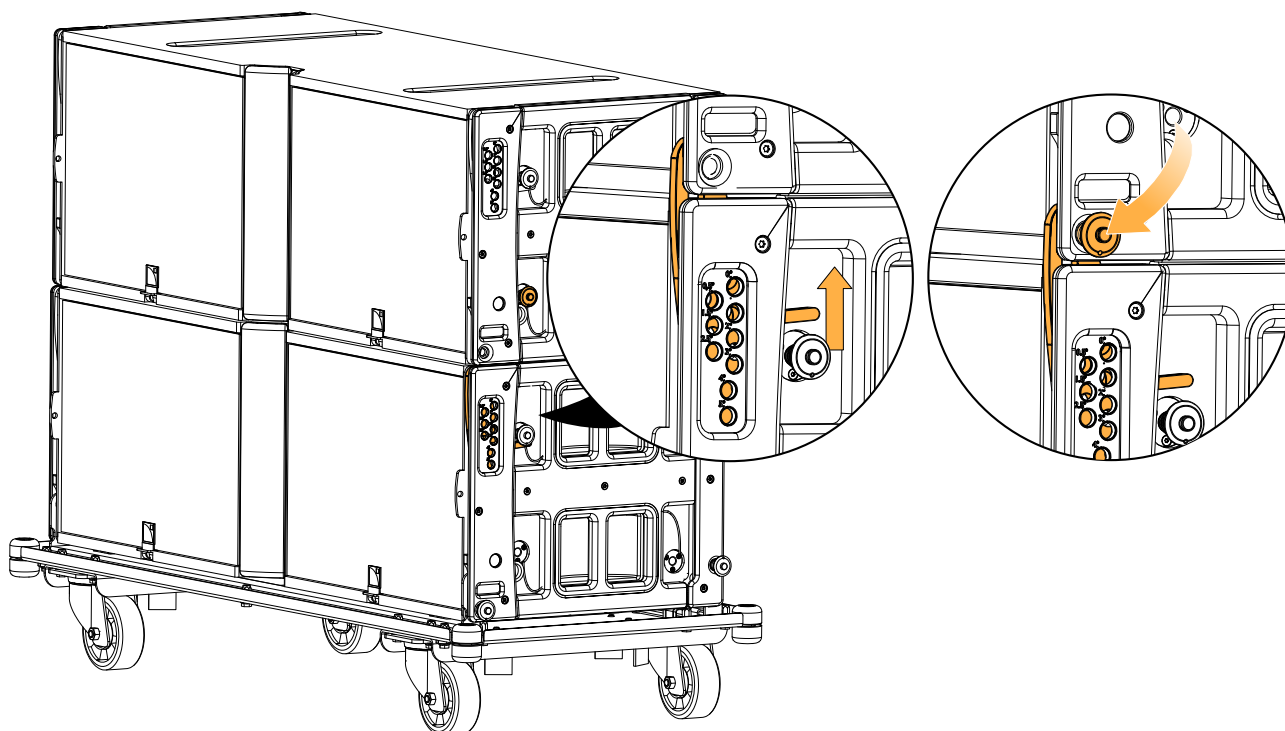
Enclosure can face either way.



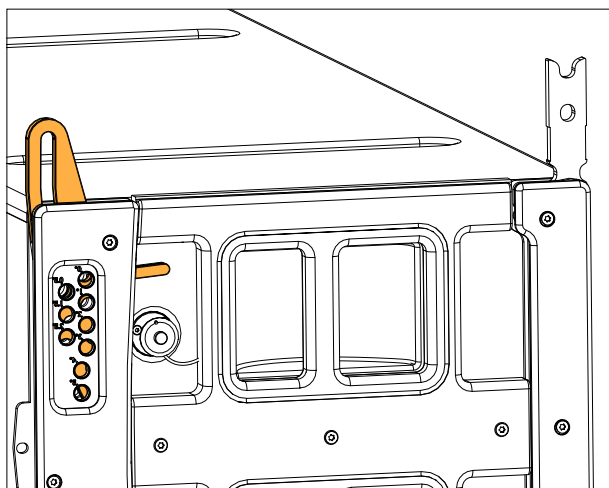
2. Position one K1 on the first one.



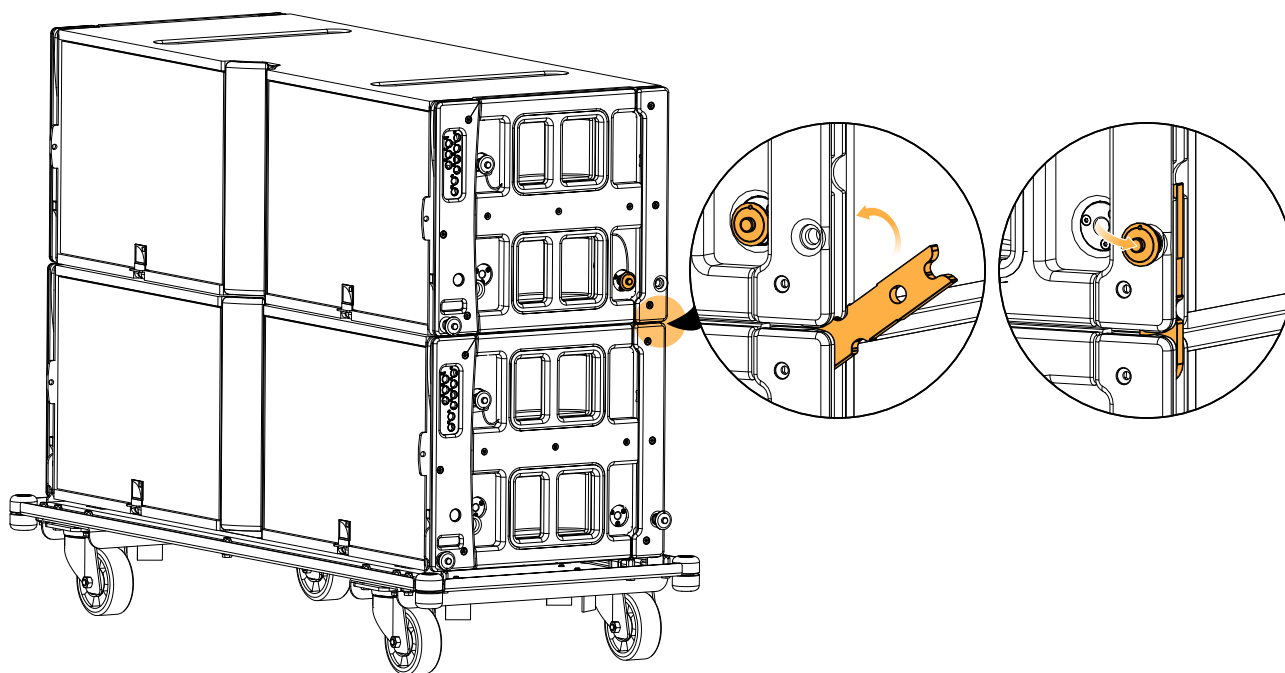
- 3.** Secure the front rigging arms using the top enclosure pins.



The front rigging arms can be deployed with some resistance, and slide correctly.



4. Secure the rear rigging arms using the top enclosure pins.

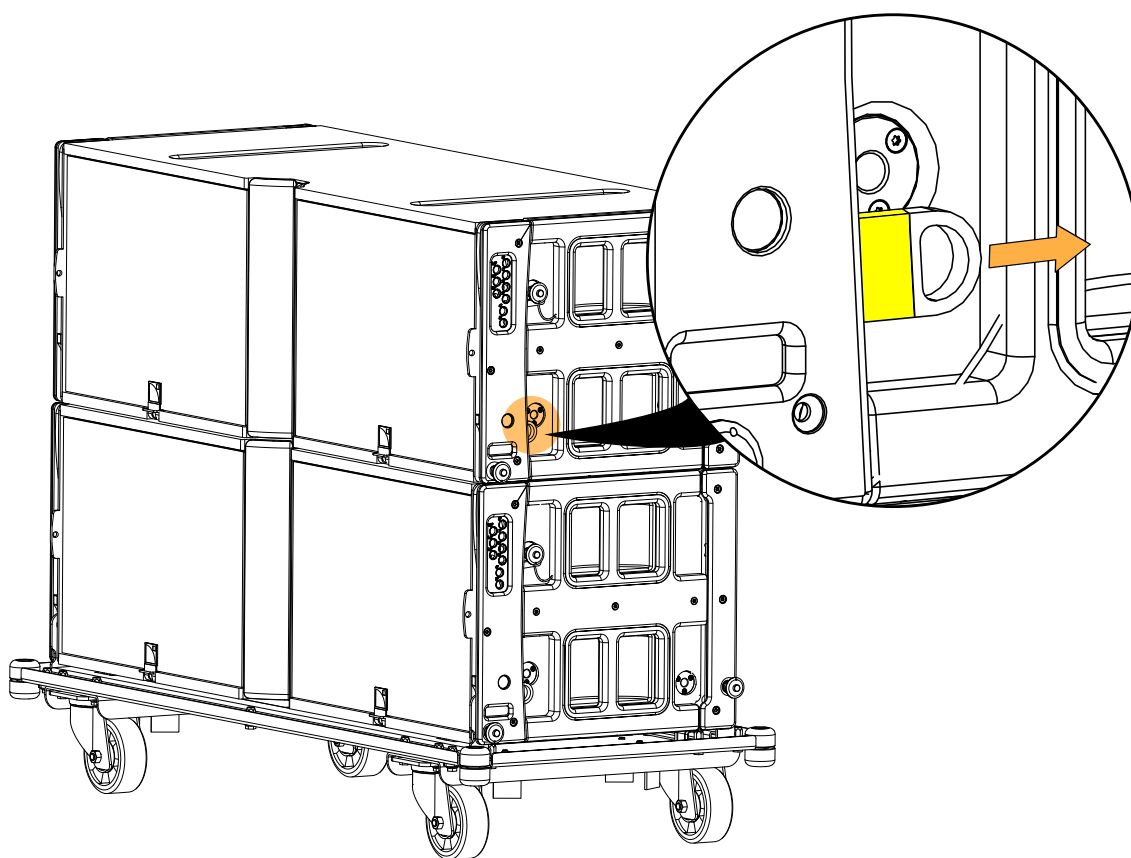


The rear rigging arms can be deployed with some resistance, and rotate correctly.

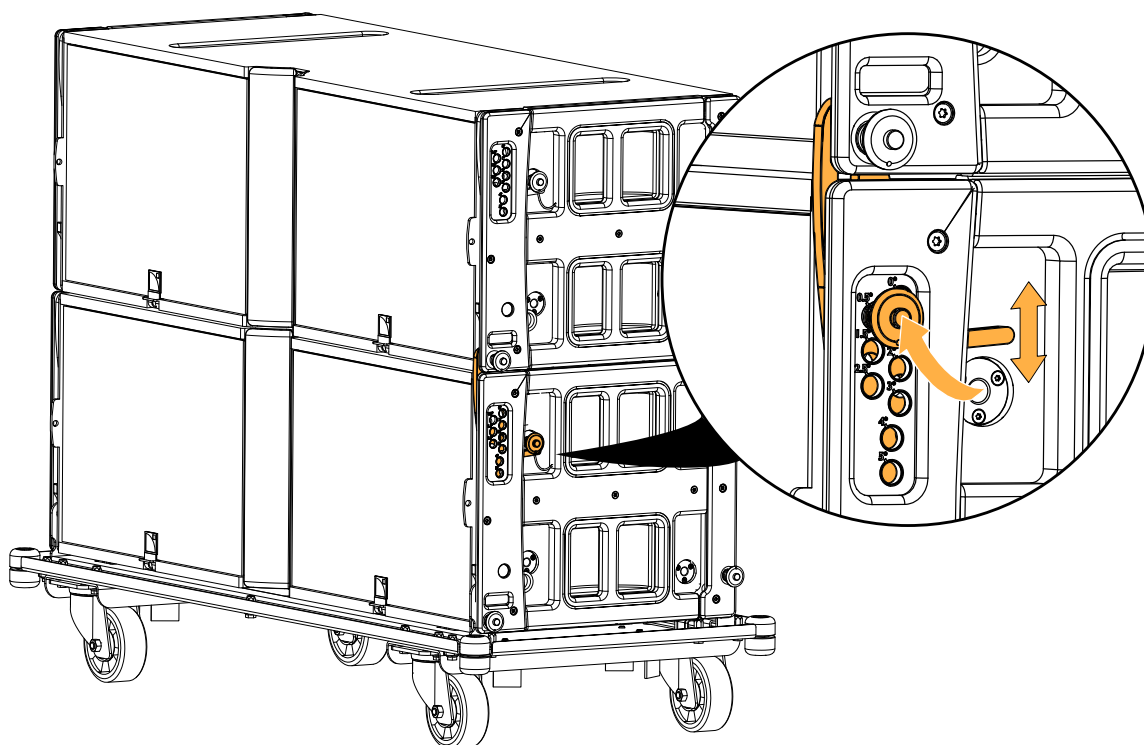


5. Make sure the automatic locking system is deactivated.
If the angle locking button is pressed, pull to disengage the latch.

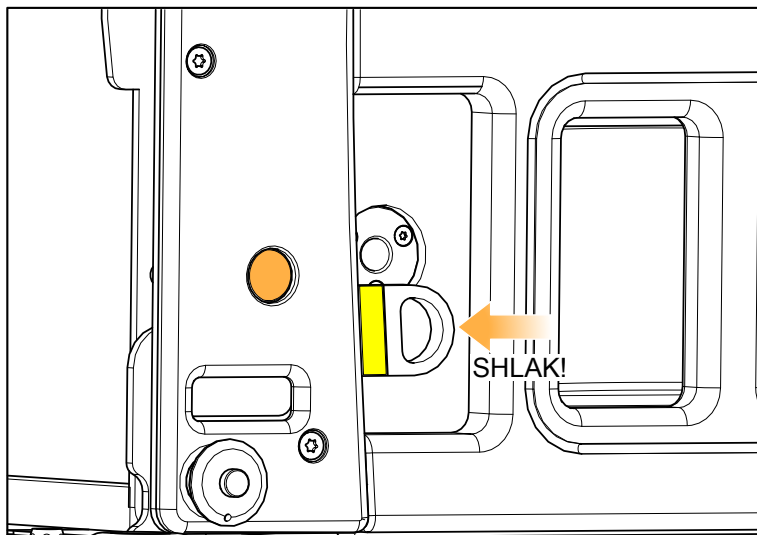
i When the automatic locking system is deactivated, **a yellow sticker is visible** on the latch.



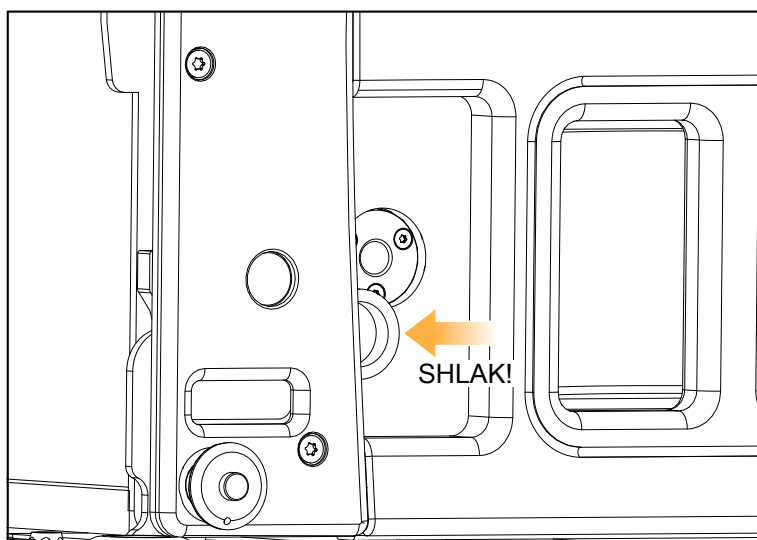
6. Secure the front rigging arms at one of the inter-enclosure holes.



- 7.** Press the button to activate the automatic locking system.
- The latch slightly retracts when the button is pushed (yellow label visible).



- 8.** Hold and lift the array to lock the inter-enclosure angles.
- The two enclosures remain attached.
 - The automatic locking system button locks and the latches engage (no yellow label visible).

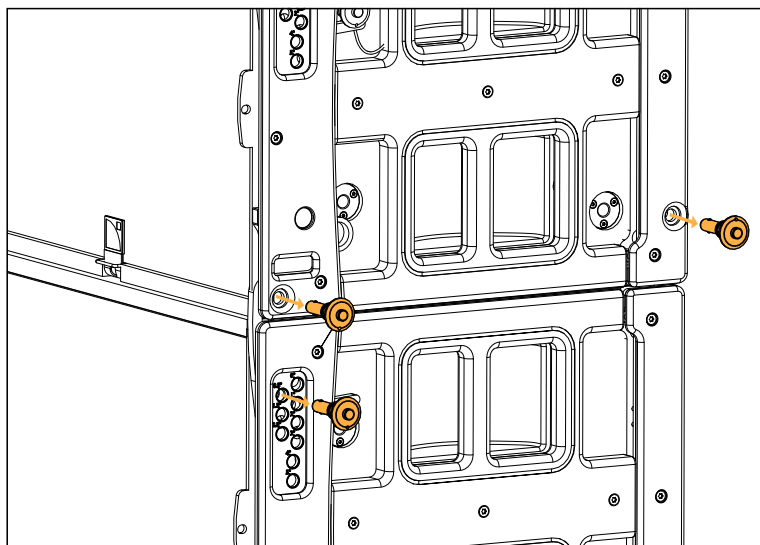


9. Disassemble the two enclosures.



Risk of trapping fingers

When disassembling the enclosures, hold the top one by the handles.



10. Repeat the procedure with the other enclosures.

Acoustical check

Enclosure check



This feature is available on:

LA12X

ENCLOSURE CHECK measures impedance at the reference frequencies for the connected loudspeaker family. The measured impedance is compared to the expected range allowing for fast detection of loudspeakers presenting circuit continuity issues.



The results can be used for preliminary diagnosis but cannot replace a comprehensive quality control.

Prerequisite



ENCLOSURE CHECK measurements can only be reliable if the following requirements are met:

Environment and temperature:

- Ambient temperature must be comprised between 0 °C / 32 °F and 40 °C / 104 °F. Ideal temperature is 20 °C / 68 °F.
- Enclosures must be at room temperature. If warm from a recent high level use or recently moved from a cold environment, let the loudspeakers reach room temperature before starting.

Enclosures:

- Enclosures must be included in the embedded factory preset library.
- Enclosures must be in nominal operating conditions:
 - Remove covers or dollies obstructing the loudspeakers or the vents.
 - Check for obvious physical damage or air leak: visually inspect the grill, gasket, cabinet, and connector plate for loose, missing or damaged parts.

Connection:

- Use only 10 m / 30 ft 4 mm² / AWG 11 speaker cables.
- Do not connect enclosures in parallel.

Amplified controllers:

- Load a preset corresponding to the connected loudspeaker's family. Presets from the user memories may be used on condition they are made of presets supported in the embedded factory preset library.

Procedure

1. Power up the amplified controller.
2. Connect the loudspeaker enclosures to the amplified controller.
3. Load a preset from or built from the embedded library corresponding to the connected loudspeaker family.
4. On the amplified controller, use the encoder wheel to select **MONITORING & INFO**. Press the OK key or the encoder wheel to validate.
5. Use the encoder wheel to select **ENCLOSURE CHECK**.



Beware of sound levels.

Although the sound pressure levels generated for the ENCLOSURE CHECK are moderate, do not stay within close proximity of the loudspeakers and consider wearing ear protection.

6. Press the OK key or the encoder wheel to launch the ENCLOSURE CHECK.

The amplified controller generates short sinusoidal signals simultaneously for each connected output.

The amplified controller displays the results for each output.

7. Depending on the displayed results, follow the instructions in the table.

result	interpretation	instructions
OK	measured impedance is within expected range	enclosure is in working order electrically
?	unsupported preset family	only supported enclosures should be tested
NC	Not Connected	if cables are connected: a. inspect the cables and connections b. go to step 8 (p.59)
NOK	measured impedance is not within expected range	a. check that all the prerequisites are met, in particular that the loaded preset corresponds to the connected speaker's family b. inspect the cables and connections c. go to step 8 (p.59)
UNDEF	measured impedance is undefined	

8. Under NC, NOK and UNDEF results, press and hold the corresponding OUT key.

The amplified controller displays:

- the tested frequencies,
- information on the measured impedance:
 - OPEN for open circuit (found in NC results),
 - SHORT for short circuit (found in NOK results), or
 - a percentage of variation from the expected range (found in NOK and UNDEF results)
- the number of operational transducers out of the total



Low variations from the expected range are acceptable: displayed percentage can be different from 0 and all transducers considered operational.

Listening test

enclosure	preset	usable bandwidth
K1	[K1]	35 Hz - 20 kHz
K2	[K2 70]	35 Hz - 20 kHz
Kara II	[KARA II 70]	55 Hz - 20 kHz
K1-SB	[K1 SB_60]	30 Hz - 80 Hz

Procedure

1. Load the preset on an LA12X amplified controller.
2. Connect a sine wave generator to the amplified controller.



Risk of hearing damage

Set a low sound level to start and use ear protection to adjust before testing.

3. Scan the bandwidth focusing on the usable range.
The sound should remain pure and free of unwanted noise.

Troubleshooting for LF/MF speakers

One or more LF/MF speaker produces distorted, buzzing, rubbing, clicking, muffled or weak sound.

Possible causes

- The screws are not tightened with the appropriate torque.
- There is an air leak in the gasket.
- There is dust on the cone.
- The cone is damaged.
- The surround is torn or delaminated.
- The voice coil or the spider is damaged.

Procedure

1. Perform the speaker disassembly procedure.
2. Visually inspect the cables and the connectors.
3. Visually inspect the speaker cone, the voice coil and the spider.

If any damage is visible, replace the speaker.

4. Carefully clean the speaker with a dry cloth.

5. Perform the reassembly procedure.

Replace the speaker gasket and the screws.

Apply the recommended torque.

6. Repeat the listening test.

If the problem persists, replace the speaker.

Troubleshooting for HF drivers

One or more HF driver produces high-frequency harmonic distortions, strange vibrations or weak sound.

Possible causes

- There are foreign particles on the air gap.
- The diaphragm is not centered correctly.
- The screws used for reassembly are too loose.
- The diaphragm is damaged.

Procedure

1. Perform the diaphragm disassembly procedure.
2. Visually inspect the diaphragm and the voice coil.
If any damage is visible, replace the diaphragm.
3. Clean the air gap thoroughly.
Use double-face adhesive tape to remove any particles.
4. Perform the diaphragm reassembly procedure.
Apply the recommended torque.
5. Repeat the listening test.
If the problem persists, replace the driver.